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Supplemental material 1. CONSORT (Consolidated Standards of Reporting Trials) and TIDieR (Template for Intervention Description and Replication) checklists

Table S1. CONSORT (Consolidated Standards of Reporting Trials) checklist

Section/Topic	Item No	Checklist item	Reported on Page No
Title and abstract			
	1a	Identification as a randomized trial in the title	0
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	0
Open science			
Trail registration	2	Name of trial registry, identifying number (with URL) and date of registration	3
Protocol and statistical analysis plan	3	Where the trial protocol and statistical analysis plan can be accessed	3
Data sharing	4	Where and how the individual de-identified participant data (including data dictionary), statistical code and any other materials can be accessed	3
Funding and conflicts of interest	5a	Sources of funding and other support (eg, supply of drugs), and role of funders in the design, conduct, analysis and reporting of the trial	20
	5b	Financial and other conflicts of interest of the manuscript authors	21
Introduction			
Background and rationale	6	Scientific background and rationale	1
Objectives	7	Specific objectives related to benefits and harms	2
Methods			
Patient and public involvement	8	Details of patient or public involvement in the design, conduct and reporting of the trial	3
Trial design	9	Description of trial design including type of trial (eg, parallel group or crossover), allocation ratio, and framework (for example, superiority, equivalence, non-inferiority or exploratory)	3, Fig. 1
Changes to trial protocol	10	Important changes to the trial after it commenced including any outcomes or analyses that were not prespecified, with reason	10, Fig.1
Trial setting	11	Settings (such as community or hospital) and locations (eg, countries or sites) where the trial was conducte	3
Eligibility criteria	12a	Eligibility criteria for participants	4, Suppl.2(Table S3)
	12b	If applicable, eligibility criteria for sites and for individuals delivering the interventions (eg, surgeons or physiotherapists)	NA
Intervention and comparator	13	Intervention and comparator with sufficient details to allow replication. If relevant, where additional materials describing the intervention and comparator (eg, intervention manual) can be accessed	3 and Ref.49
Outcomes	14	Prespecified primary and secondary outcomes, including the specific measurement variable (eg, systolic blood pressure), analysis metric (for example, change from baseline, final value, time to event), method of aggregation (eg, median, proportion), and time point for each outcome	5,6 Suppl. 3, 4 and 5
Harms	15	How harms were defined and assessed (eg, systematically or non-systematically)	NA
Sample size	16a	How sample size was determined, including all assumptions supporting the sample size calculation	Ref. 43
	16b	Explanation of any interim analyses and stopping guidelines	NA
Randomization			
Sequence generation	17a	Who generated the random allocation sequence and the method used	Ref. 43
	17b	Type of randomization and details of any restriction (eg, stratification, blocking and block size)	Ref. 43

Allocation concealment mechanism	18	Mechanism used to implement the random allocation sequence (eg, central computer/telephone; sequentially numbered, opaque, sealed containers), describing any steps to conceal the sequence until interventions were assigned	Ref. 43
Implementation	19	Whether the personnel who enrolled and those who assigned participants to the interventions had access to the random allocation sequence	Ref. 43
Blinding	20a	Who was blinded after assignment to interventions (eg, participants, care providers, outcome assessors, data analysts)	Ref. 43
	20b	If blinded, how blinding was achieved and description of the similarity of interventions	Ref. 43
Statistical methods	21a	Statistical methods used to compare groups for primary and secondary outcomes, including harms	8
	21b	Definition of who is included in each analysis (eg, all randomized participants), and in which group	8
	21c	How missing data were handled in the analysis	8
	21d	Methods for any additional analyses (eg. subgroup and sensitivity analyses), distinguishing prespecified from post hoc	8
Results			
Participant flow, including flow diagram	22a	For each group, the numbers of participants who were randomly assigned, received intended intervention, and were analyzed for the primary outcome	3, Fig. 1
	22b	For each group, losses and exclusions after randomization, together with reasons	Fig. 1
Recruitment	23a	Dates defining the periods of recruitment and follow-up for outcomes of benefits and harms	3
	23b	If relevant, why the trial ended or was stopped	NA
Intervention and comparator delivery	24a	Intervention and comparator as they were actually administered (eg, where appropriate, who delivered the intervention/ comparator, how participants adhered, whether they were delivered as intended (fidelity))	12
	24b	Concomitant care received during the trial for each group	13
Baseline data	25	A table showing baseline demographic and clinical characteristics for each group	3, Fig.1, Suppl.6(Table S10)
Numbers analyzed, outcomes and estimation	26	For each primary and secondary outcome, by group: • the number of participants included in the analysis • the number of participants with available data at the outcome time point • result for each group, and the estimated effect size and its precision (such as 95% confidence interval) • for binary outcomes, presentation of both absolute and relative effect size	10, Fig.1, Fig.2A, Fig.2B, Fig.3A, Suppl.6(Table S13, Fig.S3, Fig.S4, Fig.S5, Fig. S6)
Harms	27	All harms or unintended events in each group	NA
Ancillary analyses	28	Any other analyses performed, including subgroup and sensitivity analyses, distinguishing pre-specified from post hoc	10, Fig.2C, Fig.3B, Fig.4, Suppl.6(S11, S12)
Discussion			
Interpretation	29	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	14
Limitations	30	Trial limitations, addressing sources of potential bias, imprecision, generalisability, and, if relevant, multiplicity of analyses	19

Table S2. TIDieR (Template for Intervention Description and Replication) checklist

Item number	Item	Where located	
		Primary Paper (page)	Supplementary material
1.	**BRIEF NAME**	0	

	Provide the name or a phrase that describes the intervention.		
2.	**WHY** Describe any rationale, theory, or goal of the elements essential to the intervention.	1	
3.	**WHAT** Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	4,8	Ref.49
4.	**PROCEDURES** Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	4,8	Ref.49
5.	**WHO PROVIDED** For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	4	Ref.49
6.	**HOW** Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	4	Ref.49
7.	**WHERE** Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	4	Ref.49
8.	**WHEN and HOW MUCH** Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	4	Ref.49
9.	**TAILORING** If the intervention was planned to be personalized, titrated or adapted, then describe what, why, when, and how.	4	Ref.49
10.	**MODIFICATIONS** If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	4	Ref.49
11.	**HOW WELL** Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	13	Ref.49
12.	Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	13	Ref.49

Supplemental material 2. AGUEDA inclusion and exclusion criteria for selecting participants

Table S3. AGUEDA inclusion and exclusion criteria for selecting participants.

Inclusion criteria	Exclusion criteria
- Men and women 65-80 years old.	- Ambulatory with pain or regular use of an assisted walking device.
- Able to speak and read fluent Spanish.	- Medical contraindication for inclusion in an exercise program.
- Living in community settings during the study.	- Neurological condition (Multiple Sclerosis, Parkinson Disease, Dementia) or brain injury (traumatic or stroke).
- Reliable means of transportation.	- Current diagnosis and treatment of a DSM V Axis I or II disorder including major depression, and seeing a psychologist, therapist, or psychiatrist in the last year.
- Being physically inactive: (i) not participating in any resistance exercise programs in the last 6 months, and (ii) accumulating less than 600 METs/week of moderate-vigorous physical activity.	- History of major psychiatric illness including schizophrenia, general anxiety disorder or depression (GDS-30≥15).
- Classified as cognitively healthy according to STICS-m, MoCA and MMSE.	- Current treatment for congestive heart failure, angina, uncontrolled arrhythmia, deep venous thrombosis or another cardiovascular event.
	- Myocardial infarction, coronary artery bypass grafting, angioplasty or other cardiac condition in the last year.
	- Current or previous treatment for any type of cancer.
	- Type I Diabetes or uncontrolled Type II Diabetes defined as insulin dependent.
	- Current treatment for alcohol or substance abuse.
	- Presence of metal implants (e.g., pacemaker, stents, joint replacement) that would be MRI ineligible.
	- Claustrophobia.
	- Color blindness.
	- Diagnosis of COVID-19 with hospitalization in intensive care unit.
	- Any other consideration that interferes with the study aims and could be a risk to the participant, at the discretion of the researcher

DSM: Diagnostic and Statistical Manual of Mental Disorders, GDS: Geriatric Depression Scale, METs: Metabolic Equivalents, MMSE: Mini-Mental State Examination; MoCA: Montreal Cognitive Assessment. STICS-m: Modified Spanish Telephone Interview of Cognitive Status; MRI: Magnetic resonance imaging. COVID-19: Coronavirus disease-19

Supplementary material 3. Cognitive outcomes

1. Description of the cognitive test

Participants completed a comprehensive neuropsychological evaluation using both paper-based and computer-based assessments. All computerized cognitive tests were administered via PC or iPAD with a standard monitor and keyboard. Each test is described below.

1.1 Montreal Cognitive Assessment

The Montreal Cognitive Assessment (MoCA) is a widely used and validated screening instrument for detecting cognitive impairment (60). The MoCA involves brief assessments of short-term memory, visuospatial abilities, orientation, attention, language, and executive functions, each of which is scored separately. These scores can be summed to calculate a total score or used as individual scores for each section. Participants can receive up to 30 points on the MoCA. The Clock Drawing Test is part of the MoCA and assesses visuospatial and planning abilities. Participants were instructed to draw a clock, including placing the hands to represent a specific time (e.g., 11:10). Additionally, the MoCa delayed recall was measured to assess participants ability to remember a list of words after a short delay. If they couldn't recall the words, multiple-choice cues were provided to help them remember (maximum score 3 points).

1.2 Wechsler Adult Intelligence Scale

The Wechsler Adult Intelligence Scale (WAIS) is a paper-based test that provides a detailed evaluation of intellectual abilities by assessing skills such as reasoning, problem-solving, memory, and processing speed (137). Two subtests from the WAIS were used:

(i) The Matrix Reasoning subtest from the WAIS evaluates visuospatial reasoning and problem-solving skills. Participants were shown a series of incomplete geometric patterns and were asked to select the option that completes the logical sequence, (ii) The Block Design subtest measures visuospatial processing and abstract reasoning. Participants were shown a series of visual patterns and asked to replicate them using colored cubes within a time limit. Accuracy and response times were recorded in both tests (e.g., higher scores reflect better cognitive performance)

1.3 Trail Making Test

The Trail Making Test (TMT) is a paper-and-pencil test that measures cognitive flexibility and set-shifting (54). TMT, Part A, is a measure of psychomotor processing speed that requires participants to connect numbers displayed on a page in ascending order. TMT, Part B, measures set-shifting and requires participants to alternate between numbers and letters to connect them in ascending order. Time to completion (seconds) was measured, and reverted (multiplied by -1) with higher values reflecting better performance.

1.4 Digit Symbol Substitution Test

The Digit Symbol Substitution Test (DSST) is a paper-and-pencil test designed to assess psychomotor processing speed and basic attention (55). The DSST requires copying as many novel symbols corresponding to numbers as possible in 120 seconds. Total number of correct responses was measured. Higher scores reflect better cognitive performance.

1.5 Rey Auditory Verbal Learning Test

The Rey Auditory Verbal Learning Test (RAVLT) assesses verbal episodic memory (61). Participants are presented with a list of words and are asked to recall them. The total learning correct score is the sum of the 5 consecutive learning trials from list A. The delayed recall score measures the number of words correctly recalled from List A after a 20-minute delay. The recognition score reflects the number of correct responses when participants identify words from the original list among new words. A recognition index was measured by the number of correct responses of the recognition trial minus the number of intrusions false correct responses of the recognition trial. Higher scores reflect better cognitive performance.

1.6 Rey - Osterrieth Complex Figure Test

The Rey-Osterrieth Complex Figure Test (ROFT) evaluates visual episodic memory and visuoconstructive skills (138). Participants are first asked to copy a complex geometric figure. Later, after a 3 minute delay, they are asked to reproduce the figure from memory. The number of correct elements recalled in both trials was recorded (maximum score of 36 points). Higher scores reflect better cognitive performance.

1.7 Picture Sequence Memory Test

Picture Sequence Memory Test (PSMT) is a computer-based test from NIH Toolbox Cognition Battery (version 1.17)(59), that evaluates episodic memory. Participants were presented with a series of pictures in a specific order, and their task was to remember the order in which the pictures were shown. The number of pictures the participants

were required to order ranged from 15-picture sequences (First trial) to 18-picture sequences (Second trial). Raw scores reflect the cumulative number of adjacent pairs of pictures remembered correctly over the two trials. Higher scores reflect better cognitive performance.

1.8 Dimensional Change Card Sort Test

Dimensional Change Card Sort Test (DCCST) is a computer-based test from NIH Toolbox Cognition Battery (version 1.17) which assesses cognitive flexibility and the ability to shift attention between different dimensions or rules (59). Participants were trained on one rule and then required to shift their attention and adapt to a new sorting rule during the shift phase. In test trials, participants sorted cards based on the current rule presented on the screen, which could switch between shape and color. Accuracy and response times were recorded (e.g., higher scores reflect better cognitive performance)

1.9 List Sorting Working Memory Test

List Sorting Working Memory test (LSWMT) is a computer-based test from NIH Toolbox Cognition Battery (version 1.17) that assesses working memory (65). Participants were asked to recall and sequence different stimuli by size that were presented both visually and via audio across two trials. In 1-list trials, all stimuli were from the same category (foods or animals). In 2-list trials, the stimuli were from two different categories (foods and animals). The participant was asked to sequence the food stimuli followed by the animal stimuli. The total number of correct items across the 1-list and 2-list conditions (maximum score of 28) were recorded.

1.10 Flanker Test

The Flanker Test is an arrow version of the computer-based test from NIH Toolbox Cognition Battery (version 1.17) that measures inhibition (59). During this test, participants are presented with a series of five arrows displayed in a row on the screen, each pointing either left or right. Participants must respond as quickly as possible to the direction of the middle arrow (left or right) using the corresponding arrow keys on the keyboard, while ignoring the external arrows.

Accuracy and response times were recorded. Higher scores reflect better cognitive performance.

1.11 Stroop Test

The Stroop Test is a computerized measure of inhibitory control from E-prime v2.0 (Psychological Software Tools, Pittsburgh, PA, USA). Words were presented on the screen one at a time. The words are colour words ‘green’, ‘blue’, ‘red’, ‘yellow’ that are printed in these different coloured inks. The participant is required to press a key

with the colour indicated on for the ink colour, inhibiting the written word presented. There were 36 practice trials (22 congruent trials; written word and ink colour the same, 14 incongruent trials ;written word and ink colour different) and 72 experimental trials (36 congruent, 36 incongruent). Accuracy and reaction times were recorded. Higher scores reflect better cognitive performance.

1.12 Spatial Working Memory Test

Spatial Working Memory Test (SWMT) is a computer-based test that assesses an individual's ability to temporarily store and manipulate spatial information from E-prime v2.0 (Psychological Software Tools, Pittsburgh, PA, USA) (139). The participants were asked to stare at a crosshair in the middle of the screen. Afterwards, 2, 3 or 4 black dots appear at random locations on the screen and disappear. Then, a red dot appears on the screen and the participant must respond if the dot either is or not placed in the same place as one of the black dots that appeared before as fast as possible. Accuracy and reaction times were recorded. Higher scores reflect better cognitive performance.

1.13 Task Switching Test

The Task-Switching Test computer test assessed the ability to flexibly switch the focus of attention between multiple task sets from E-prime v2.0 (Psychological Software Tools, Pittsburgh, PA, USA). Participants had to switch between judging whether a number was odd or even and judging whether it was higher or lower than 5. The eligible numbers were 1, 2, 3, 4, 6, 7, 8, and 9, and were presented individually for 1500 ms in the center of the computer screen, enclosed within either a circle or a square, with the constraint that the same number did not appear twice in succession. The shape of the frame determined the task rule: if the number was inside a circle, participants had to decide whether it was higher or lower than 5, whereas if the number was inside a square, they had to determine whether it was odd or even. Participants responded using their left hand for the high/low task (‘x’ key for high, ‘z’ key for low) and their right hand for the odd/even task (‘n’ key for odd, ‘m’ key for even). The test consisted of three blocks: a first block (24 trials) where all numbers appeared inside circles, requiring only the high/low classification; a second block (24 trials) where all numbers appeared inside squares, requiring only the odd/even classification; and a third mixed-task block (120 trials) where numbers appeared inside either circles or squares, requiring participants to switch between tasks based on the shape of the frame. In the switching block, trials were categorized as repeat trials (where the task remained the same as in the previous trial) or switch trials (where the task changed from the previous trial). The switch cost was calculated as the difference in reaction times

between switch and repeat trials, reflecting the cognitive effort required to adapt to a new task rule. Reaction times, accuracy rates, and switch costs were recorded to assess cognitive flexibility and task-switching efficiency.

1.14 N-back

The N-back computer test is a measure of visual working memory from E-prime v2.0 (Psychological Software Tools, Pittsburgh, PA, USA) (66), which was completed by participants during their MRI scan. On the 1-back condition, participants were instructed to indicate whether the letter on the screen matched the letter that was previously displayed. The 2-back condition requires a greater cognitive load and instructed participants to indicate whether the letter on the screen matched the letter that was displayed two trials previously. Accuracy and reaction time for both conditions were recorded.

2. Executive function score

2.1 Creation of composite executive function score

2.1.1 Methods

Z-standardization was used to ensure that outcomes with different response scales were on a standardized and comparable metric. Inverse variables were used when needed so a higher value indicated better performance. Firstly, a preliminary data checking procedure was conducted: Means, minimums and maximums of raw grounded outcomes (i.e., reaction times and accuracies) of each cognitive indicator were inspected. Pearson correlations were performed to test how the cognitive indicators related to each other.

Secondly, analyses were conducted to ensure that assumptions of normality and linearity. Bartlett's Sphericity Test assessed the probability that at least some of the variables were significantly correlated and the Kaiser-Meyer-Olkin (KMO) statistic assessed the factorability of the data using the R package performance (140). The KMO statistic, ranging from 0 to 1, predicted whether the data were likely to factor well given the correlations among the variables. Using Kaiser's guidelines (141) a cutoff of KMO $\geq$ .60 was used.

The CFA was conducted using the "*lavaan"* package in R (142) with maximum likelihood estimation. We tested the fit of several models that differed in the following ways for creating the executive function score: i) whether model fit improved with three cognitive domains factors, ii) whether model fit improved with a second-order general cognitive factor and iii) whether the model fit was better with a first-order factor of executive function

Firstly, guided on available evidence and theoretical positions of executive function, a model that hypothesized three factors was tested. Specifically, a model with three cognitive domains factors: 1) Working memory derived

from SWMT, PSMT and LSMT, 2) Inhibitory control from Flanker test and Stroop test and 3) Cognitive flexibility from Task switching, DCCST, DSST and TMT. Secondly, because of a bad fit of the previous model we performed a two-factor model and a second-order one dropping indicators with factor loadings lower than 0.45, specifically, with 1) working memory derived from SWMT, PSMT and LSMT and 2) Cognitive flexibility from DCCST, DSST, TMT and Flanker Test.

Finally, the first-order CFA with a general executive function factor was the one that best fit, which added task-specific covariance that allowed the scores within a task to correlate with one another, with loadings freely estimated (143). The latent scale was identified by fixing the loading of a reference indicator to one. The remaining pattern coefficients, factor variances, and factor covariances were freely estimated (144). Each criterion was unidimensional and loaded on only one factor. See Table S4 for a list of measures and outcomes analyzed.

Five different goodness-of-fit statistics were used to assess model fit using established cut-offs, including the $\chi 2$ (p-value $\geq$ .05) (145), the comparative fit index (CFI) (> .95), the root mean square error of approximation (RMSEA) (< .08) (146), the Standardized Root Mean Square Residual (SRMR) (< .08) (147,148), and indicators with factor loadings less than 0.45 were removed to improve model fit, as it has been suggested that a cut-off of 0.45 indicates good (30%) overlapping variance (149).

Table S4. Executive function Measures and conditions included in the hypothesized and final factor analysis models

Measure	Outcome	Hypothesized	Final
Trail Making test (TMT)	Interference score (Time part B – Time part A) (sec)	✔	✔
Digit Symbol Substitution test (DSST)	Total correct score (n)	✔	✔
Dimensional Change Card Sort Test (DCCST)	Inverse efficiency score (switch trials - high load) (rt/acc)	✔	✔
Spatial Working Memory Test (SWMT)	Inverse efficiency score of all trials (rt/acc)	✔	✔
Picture Sequence Memory Test (PSMT)	Theta Score*	✔	
List Sorting Working Memory Test (LSWMT)	Total correct score (n)	✔	
Flanker Test	Inverse efficiency score of Incongruent trials (rt/acc)	✔	
Stroop Test	Inverse efficiency score of Incongruent trials (rt/acc)	✔	
Task Switching Test	Global cost score (switch trials - non-switch trials)	✔	

Abbreviations: acc, accuracy; n, number; sec, seconds; rt, reaction time. *Theta score is a statistical measure that reflects a participant's relative performance, similar to a z-score, which is a measure that indicates how far a participant's score is from the average (mean) of a normative group, in terms of standard deviations.

2.1.2 Results

2.1.2.1 Data preprocessing and winsorization

The 90 participants randomized were reviewed for consistency by checking the minimum and maximum values of the raw outcomes and evaluating the biological plausibility of the data to identify any implausible values. For sensitivity analyses, 27 outliers, defined as values exceeding four standard deviation from the mean, were

winsorized by replacing them with the nearest non-extreme value (See Table S5). However, winsorization was not applied in the main analyses.

Table S5. Cognitive measure, outcome and the number of participants winsorized

Cognitive measure	Outcome (unit)	Number of participants winsorized
Rey Auditory Verbal Learning Test (RAVLT)	Recognition - Number of correct responses of the recognition trial (n)	2
Rey - Osterrieth Complex Figure Test (ROFT)	Raw score copy (n)	1
Trail Making Test (TMT)	Time part A (sec)	3
	Time part B (sec)	5
	Inverse efficiency score of all trials (rt/acc)	5
Dimensional Change Card Sort Test (DCCST)	Computed score* (n)	4
Dimensional Change Card Sort Test (DCCST)	Inverse efficiency score (switch trials - high load) (rt/acc)	3
Spatial Working Memory Test (SWMT)	Inverse efficiency score of all trials (rt/acc)	4

Abbreviations: acc, accuracy; n, number; sec, seconds; rt, reaction time. *Theta score is a statistical measure that reflects a participant's relative performance, similar to a z-score, which is a measure that indicates how far a participant's score is from the average (mean) of a normative group, in terms of standard deviations. Winsorization was applied only in the sensitivity analyses and not in the main analyses.

2.1.2.2 Missing data

Of the 90 randomized participants, 87 successfully completed all the baseline cognitive assessments included in the executive function scores and other domains. 69 successfully completed all the cognitive assessments at both baseline and at 24-weeks. The number of missing data points from the original randomized sample at 24-weeks were as follows: for composite executive function score, attentional/inhibitory control, episodic memory, and processing speed, the RE had n = 3 missing cases, while the control group had n = 6; for visuospatial processing and working memory, the RE had n = 3, and the control group had n = 7. A chi-square of the completion ratio between groups was not significant (all $p > 0.43$), suggesting the proportion of missing data did not differ by group assignment. Moreover, a comparison of the subsample who did not successfully complete the 24-week cognitive assessment to the sample who completed both baseline and 24-week cognitive assessments revealed no differences in sex and education (all $p > 0.05$). However, age was significantly related to missing data ($p = 0.019$), indicating that older participants were more likely to have missing follow-up cognitive test data. Based on these associations,

our primary regression models were adjusted for the effect of age, assuming that missing data were missing at random (MAR), which includes covariate-dependent missing completely at random (MCAR) as a special case.

Table S6. Cognitive measure, missing data and chi-square

	Missing data		All chi-square	Sub-group chi-square		
	Exercise	Control	p-value	Sex	Age	Education
Composite executive function score and other domains	3	6	0.439	0.656	0.019	0.840

Analysis of Chi-square were done with the lower n value.

2.1.2.3 Pearson correlations between executive function measures.

Correlations between the executive function indicators are presented in **Supplementary material 3, Figure S1**. All other scores were significantly correlated with each other (ranging between r = 0.15, and r = 0.55) but the task switching task was negatively correlated.

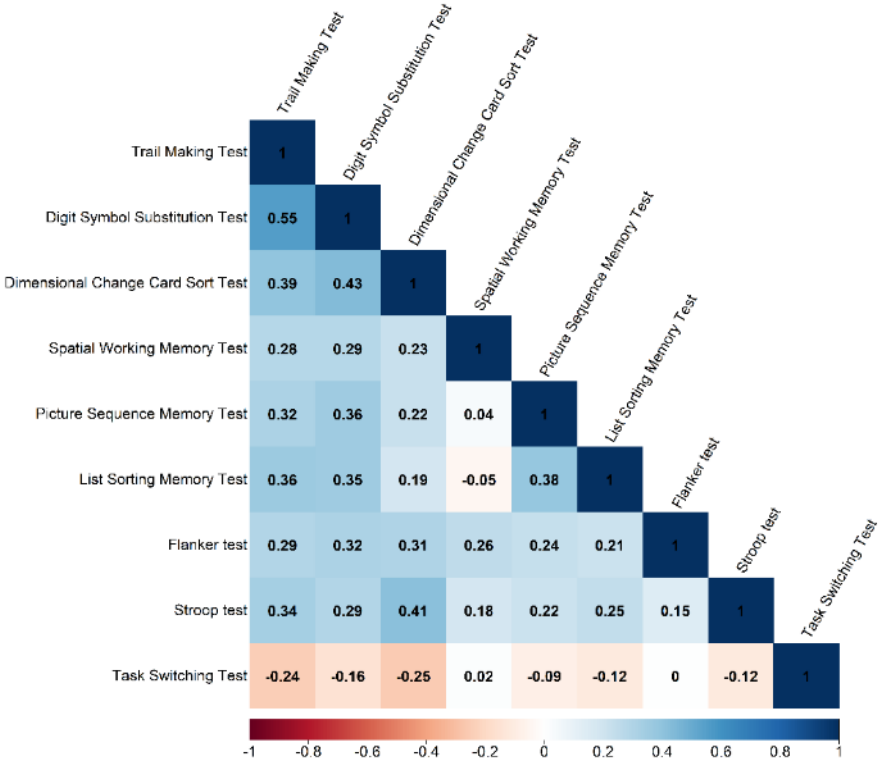


Figure S1. Pearson correlations between executive function measures. Color bar represents the strength of the correlation as r values. The darker blue the color, the stronger the positive correlation between variables, and the darker red the color, the stronger the negative correlation between variables. Blank spaces represent non – significant correlations. Trail Making Test, Dimensional Change Card Sort Test, Spatial Working Memory Test, Flanker test, Stroop test and Task Switching Test are reverted so that they can be interpreted in the same fashion as the rest of the outcomes (i.e., higher score indicates better performance).

2.1.2.4 Bartlett's test of sphericity.

Bartlett's test of sphericity was significant ($c^2(45) = 151.1341$, $p < 0.001$). The KMO measure of sampling adequacy indicated that the strength of the relationship among variables was high (KMO = 0.62) and, thus, it was acceptable to proceed with subsequent analyses.

2.1.2.5 CFA models for executive function score

The fit of the three models for creating the executive function score is detailed below: (i) The three-order factor model did not satisfy the model fit criteria (CFI = 0.825, TLI= 0.738, RMSEA = 0.096, SRMR = 0.077, $c^2(24)= 0.009$, $p < 0.001$). A negative variance on working memory was found. Due to this lower fit, a series of alternative CFA were examined, as described in the method section. (ii) The hypothesis of a model with only second-order factor did not satisfy the model fit criteria (CFI = 0.707, TLI= 0.590, RMSEA = 0.145, SRMR = 0.284, $c^2(15) = 0.000$). It was also replicated in a second-order model, but did not fit neither (CFI = 0.885, TLI= 0.815, RMSEA = 0.097, SRMR = 0.071, $c^2(13) = 0.032$).

(iii) Finally, the best-fit-model was composed of one first-order factor, SWMT, DCCST, DSST and TMT (CFI = 0.975, TLI= 0.925, RMSEA = 0.095, SRMR = 0.042, c^2 (2) =0.165, p = 0.000). **Supplementary material 3, Figure S2** presents the latent factor constructs of the final model with standardized factor loadings on the paths. All loadings were statistically significant (all $p<0.001$), and all measures had loadings >0.45.

The post hoc power analysis revealed that a sample of N = 90 is associated with power larger than > 99.99% to reject a bad-fit model (df = 28) with an amount of misspecification corresponding to RMSEA = 0.08 and alpha = 0.05.

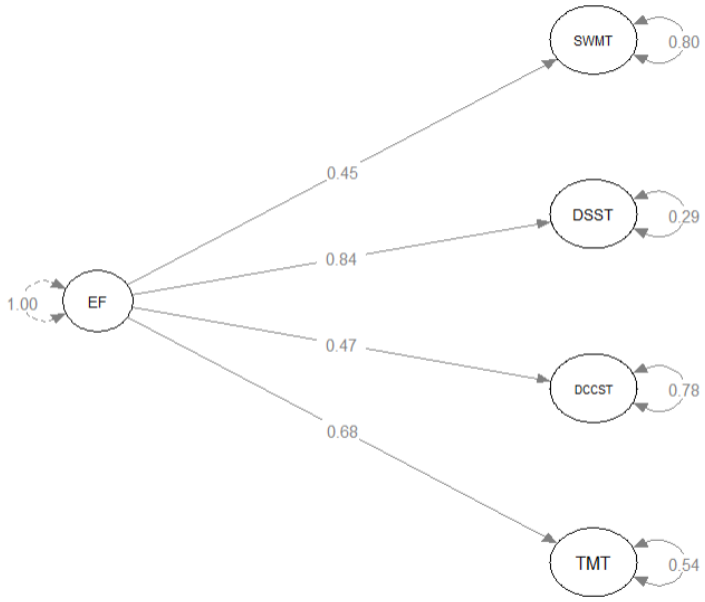


Figure S2. First-order factor analysis model. Factor loadings (numbers in the middle) represent the strength of the relationship between each test and the general executive function (EF) score. Higher loadings indicate a stronger association. The values next to each test represent the variance explained by the factor for each test. Abbreviations: TMT, Trail Making Test; DCCST, Dimensional Change Card Sort Test; SWMT, Spatial Working Memory Test; DSST, Digit Symbol Substitution Test.

2.1.2.6 Summary of the executive function composite score

The executive function composite score included the following sub-tests: SWM (43), DCCST (44), DSST (45) and TMT (46). The cognitive measure, outcome, range and mean are described in **Table S7**.

Table S7. Summary table with cognitive measure, outcome, range ad mean of the composite executive function score and the variables included in the score.

Measure	Outcome (unit)	Range	Mean (SD)
Composite executive function score	Overall z-score	-3.38 - 1.55	-0.01 (1.00)
Sub-test included in the composite executive function score			
Trail Making Test (TMT)	Interference Score (Time part B – Time part A) (sec)	-17.44 - 432.61	72.64 (74.1)
Digit Symbol Substitution Test (DSST)	Total correct score (n)	18 - 74	44.54 (13.15)
Dimensional Change Card Sort Test (DCCST)	Inverse efficiency score (switch trials - high load) (rt/acc)	0.01 - 0.07	0.01 (0.01)
Spatial Working Memory Test (SWMT)	Inverse efficiency score of all trials (rt/acc)	8.01 - 48.33	15.29 (7.35)

Abbreviations: acc, accuracy; n, number; sec, seconds; rt, reaction time. The Trail Making Test, Dimensional Change Card Sort Test, and Spatial Working Memory Test were reverted in the analyses, meaning that higher values indicate better performance. However, the data presented in the table are in their original (non-reverted) form. Baseline data are presented.

3. Cognitive domains

Cognitive domains were established by replicating a large-scale RCT with similar participant's characteristics, cognitive tests and aims, but with a considerable larger sample size (n=648) to run a multiple-factor CFA model

(35). Five cognitive domains were created: attentional/inhibitory control, episodic memory, processing speed, visuospatial processing and working memory. **Table S8** shows a summary of the cognitive measure included in each domain, along with their outcomes, ranges, and mean (SD) scores.

Table S8. Measure, outcome, range and mean of the domains included.

Domain	Cognitive measure	Outcome (unit)	Range	Mean (SD)
Attentional/inhibitory Control	Trail Making Test (TMT)	Time part B (sec)	40.02 - 500.84	122.83 (85.6)
	Dimensional Change Card Sort Test (DCCST)	Computed score* (n)	2.13 - 8.56	6.96 (1.21)
	Stroop Test	Incongruent raw score (rt)	689.13 - 1364.34	970.84 (160.09)
	Flanker Test	Computed score* (n)	4.75 - 8.57	7.07 (0.82)
Episodic memory	MoCA Delayed Recall	Multiple choice cue (n)	0.00 - 5.00	3.14 (1.25)
	Picture Sequence Memory Test (PSMT)	Raw score (n)	2.00 - 28.00	9.23 (5.53)
	Rey Auditory Verbal Learning Test (RAVLT)	Total recall raw score- Sum of the 5 consecutive learning trials from List A (n)	26.00 - 65.00	44.46 (8.36)
		Total delayed recall correct raw score - Sum of words of List A after 20min (n)	4.00 - 15.00	9.04 (2.54)
		Recognition - Number of correct responses of the recognition trial - Number of intrusions false correct responses of the recognition trial) (n)	8.00 - 15.00	14.03 (1.47)
	Rey - Osterrieth Complex Figure Test (ROFT)	Raw score copy (n)	11.50 - 36.00	32.66 (4.64)
		Delayed raw score after 3 min period – Total of points (n)	1.00 - 30.50	15.66 (7.19)
Processing Speed	Digit Symbol Substitution Test (DSST)	Total correct raw score (n)	18.00 - 74.00	44.54 (13.15)
	Trail Making Test (TMT)	Time part A (sec)	21.28 - 218.63	50.19 (23.94)
Visuospatial processing	Wechsler Adult Intelligence Scale	Block design (acc)	8.00 - 40.00	23.06 (7.18)
		Matrix reasoning (acc)	2.00 - 19.00	9.82 (4.67)
	MoCA Clock Draw	Total Score (n)	1.00 - 3.00	2.60 (0.6)
Working Memory	Spatial Working Memory Test (SWMT)	3-item (acc)	20.00 - 95.00	68.53 (15.74)
		4-item (acc)	28.00 - 90.00	67.38 (13.62)
	List Sorting Working Memory (LSWMT)	Total correct raw score (n)	8.00 - 20.00	14.88 (2.17)
	N-back	2-back (acc)	11.00 - 96.00	64.31 (22.51)

Abbreviations: acc, accuracy; min,minutes; n, number; sec, seconds; rt, reaction time. Computed Score combines accuracy and reaction time (0-10). If accuracy ≤80%, score equals accuracy; if accuracy >80%, score is a combination. The Trail Making Test and Stroop Test variables were reverted in the analyses, meaning that higher values indicate better performance. However, the data presented in the table are in their original (non-reverted) form. Baseline data are presented.

Supplementary material 4. Moderation outcomes

1. Moderators criteria

To evaluate the moderating effects of individual characteristics, participants were categorized based on various key variables. The following sections outline the criteria and methods used to define each category.

1.1 Sex

Sex was categorized as male or female based on self-reported information regarding the sex assigned at birth.

1.2 Age

Age was dichotomized as youngers (<72 years old) or orders (≥72 years old), based on the median of study's age range used for randomization (65-80 years).

1.3 Educational level

Educational level was grouped as low (≤12 years) or high (>12 years), based on the completion of high school or equivalent as a threshold.

1.4 APOE4 carriership

The genotyping of selected APOE SNPs (rs7412 and rs429358), which determine the presence of ε2, ε3, and ε4 alleles, was performed at GENYO using KASPar® assays (LGC Genomics, Hoddesdon, UK) following the manufacturer's protocol. Participants were classified as either APOE4 carriers (carrying at least one APOE ε4 allele) or non-carriers.

1.5 Amyloid burden

Amyloid burden was assessed using positron emission tomography (PET) imaging and quantified with the Centiloid (CL) scale, which standardizes amyloid measurements across different methods. A cutoff of 12 CL was used to classify participants as amyloid-negative (<12 CL) or amyloid-positive (≥12 CL), based on established thresholds for identifying elevated amyloid burden.

1.6 Comorbidities

Participants were categorized into low (<3) or high (≥3–4) comorbidity group based on the total number of conditions reported in their self-reported medical history, including diagnoses or medications related to hypertension, diabetes, heart disease, obesity, or cholesterol.

1.7 Baseline cognitive performance

Baseline cognitive performance for EF and other cognitive domains, attentional/inhibitory control, episodic memory, processing speed, visuospatial processing and working memory, was assessed using pre-intervention data. Participants were categorized into two groups: low (<median) and high (≥median), based on the median scores calculated for each domain.

1.8 Subjective cognitive decline

The subjective cognitive decline was measured with the subjective cognitive decline scale (58), utilizing four questions related to self-experience. The questions were 1) Do you have more trouble remembering things that have happened recently? 2) Are you worse at remembering where belongings are kept? 3) Do you have trouble recalling conversations a few days later? 4) Do you have more trouble remembering appointments and social arrangements? All questions had the same options to answer with specific punctuation "No, not much worse (0 points)"; "A bit worse (1 point)"; "Yes, a lot worse (2 points)". Baseline cognitive decline was categorized as low (<median) or high (≥median) based on the sum of the four answers score from 0 to 8 points (59).

Supplementary material 5. Physical condition parameters.

1. Overview of the muscular strength, physical function and cardiorespiratory fitness outcomes

Additional outcomes related to physical condition parameters were divided into three categories: muscular strength, physical function, and cardiorespiratory fitness. Physical outcomes were assessed by using the Short Physical Performance Battery (SPPB), the Senior Fitness Test (SFT) and individual tests such as the handgrip test, the isokinetic tests or the 2-km walking test. A summary table detailing the measures, tests, and outcomes is presented in **Table S9**.

Table S9. Additional outcomes related to physical condition parameters: muscular strength, physical function, and cardiorespiratory fitness

Domain	Measure (battery)	Outcome (unit)	Range	Mean (SD)
Muscular strength	Arm Curl (SFT)	Repetitions (n/30sec)	8.00 - 24.00	16.00 (3.77)
Muscular strength	30-second sit-to-stand test (SFT)	Repetitions (n/30 sec)	8.00 - 22.00	13.98 (2.97)
Muscular strength	5-times sit-to-stand test (SFT)	Time (sec)	5.16 - 16.76	9.20 (2.24)
Muscular strength	Handgrip test	Weight (Kg)	12.65 - 48.65	28.03 (9.67)
Muscular strength	Isokinetic muscular strength; Elbow extension	Newton meters (Nm)	16.00 - 247.50	89.90 (34.33)
Muscular strength	Isokinetic muscular strength: Knee extension	Newton meters (Nm)	30.00 - 199.00	95.89 (33.02)
Physical function	2-minute step test (SFT)	Repetitions (n/2min)	38.00 - 170.00	70.83 (19.18)
Physical function	Up and go test (SFT)	Time (s)	3.95 - 10.41	5.81 (1.13)
Cardiorespiratory fitness	6-minute walk test (SFT)	Distance (m)	318.80 - 693.70	477.84 (73.33)

Cardiorespiratory fitness	2 km walking test	Time (min)	15.27 - 35.67	23.29 (3.75)

Abbreviations: n, number; y, years; SD, standard deviation; sec, seconds; min, minutes; acc, accuracy; cm, centimeters; m, meters; N, Newtons; SFT, Senior Fitness Test; SPPB, Short Physical Performance Battery.
The 5-times sit-to-stand test (SFT), Up and go test (SFT) and 2 km walking test were reverted in the analyses, meaning that higher values indicate better performance. However, the data presented in the table are in their original (non-reverted) form. Baseline data are presented.

2. Muscular strength outcomes

2.1 Arm Curl

The upper body muscular strength of participants was assessed using the arm curl test which is included in the SFT battery. Participants initiated the test from a seated position and, with a dumbbell (i.e., 5 pounds for females and 8 pounds for males), flexed and extended their dominant arm for a duration of 30 seconds upon the evaluator's signal. An explanatory demonstration of the test was given by the evaluator before its initiation. The test was carried out twice, and the highest score (repetitions) obtained was used as an upper body muscular strength indicator. Higher values indicate better performance.

2.2 30-seconds sit-to-stand test

The lower body strength of participants was evaluated through the 30-seconds sit-to-stand test, which is included in the SFT battery. Participants start the test from a seated position, crossing their hands over their chests, and were instructed to rise from the chair and sit down repeatedly for a continuous 30-second period. A demonstration of the test was provided by the evaluator prior to start. The test was conducted and recorded twice, but only the highest score (repetitions) was used as a lower body muscular strength indicator. Higher values indicate better performance.

2.3 5-times sit-to-stand test

The 5-times sit-to-stand test, included on the Short Physical Performance Battery, was employed to evaluate lower body strength. At the direction of the evaluator, participants, situated in a seated position with their hands crossed on their chests, were required to perform a sequence of standing up and sitting down from the chair for a total of five repetitions. Prior to the test, the evaluator elucidated the procedure through a demonstration. The test was executed twice, with both trials measured and recorded. The best time to completion (seconds) the five repetitions was used, and reverted (multiplied by -1) with higher values reflecting better performance.

2.4 Handgrip strength test

Isometric strength was measured using the handgrip test. Upon the evaluator's signal, participants grasped a dynamometer (Takei TKK 5401) tightly with their hand, exerting maximal force. The test was administered twice for each hand. Two scores of each hand were achieved in each test by the evaluator. Subsequently, the highest result for each hand was taken into account, and the average of the two hands was calculated to ascertain the isometric strength value in kg. Higher values indicate better performance

2.5 Isokinetic muscular strength; Elbow extension

Gymmex Iso-2 dynamometer (EASYTRCH s.r.l., Italy) was used to perform the Isokinetic Strength Test for upper limbs (elbow extension). The participant's initial position was as follows: they sat with their back fully against the back of the seat, holding the grip at the end with their non-performing hand. Their elbow was positioned at about a 130° angle to the horizontal, at chest height. Before starting the exercise, the participant received detailed instructions and completed a warm-up of 10 slow, controlled repetitions. For the exercise itself, the participant extended their elbow to a 180° position five times, applying maximum possible force each time. This was done sequentially with each arm. The score was obtained from the average of the five maximum-force knee extensions for both arms, expressed in Newton meters (Nm) as peak torque. Higher values indicate better performance.

2.6 Isokinetic muscular strength ; Knee extension

This test was performed using the same isokinetic device that was used for elbow extension, Gymmex Iso-2 dynamometer (EASYTRCH s.r.l., Italy). The participant's initial position was as follows: the knee was bent at approximately a 90° angle to the horizontal, with the back fully supported against the seat, and both hands holding the grips at each end. Before starting the exercise, the participant received specific instructions and completed a warm-up. The exercise involved extending the knee to 180° a total of five times, applying maximum possible force each time. This was done sequentially with each leg. The score was obtained from the average of the five maximum-force knee extensions for both legs, expressed in Newton meters (Nm) as peak torque. Higher values indicate better performance

3. Physical function outcomes

3.1 2-minute step test

The 2-minute step test was employed to assess the physical function which is included in the SFT battery. Participants commenced the test by standing in front of a wall. The evaluator measured and marked the hip height

on the wall. During the test, participants were required to alternately lift their knees above the marked height for a duration of two minutes. The evaluator provided a demonstration of the task and initiated the test by counting the number of times the participant's right knee exceeded the marked height on the wall. The number of repetitions in 2 minutes was used as indicator of this test and higher values indicate better performance.

3.2 Up and go test

The up and go test, which is included in the SFT battery, was conducted to assess walking speed, balance, and agility . Participants initiated the test from a seated position, while the evaluator positioned a cone at 2.5 meters in front of the participant. In response to the evaluator's cue, participants stood up, navigated around the cone, and returned to a seated position. The evaluator provided a demonstration to clarify the task and then timed the participant's test completion, emphasizing in the execution without resorting to running. The test was administered twice, and the shortest time recorded (In seconds) was used as the indicator of this test. The score was reverted (multiplied by -1) so higher values indicate better performance.

3.3 Short-distance timed walk test

The objective of this test was to assess participants' walking ability from the SFT battery. The test involved measuring the time taken to walk 4 meters in an unobstructed space. Participants were instructed to walk at a regular pace. Cones marked the start and end lines. Upon the evaluator's signal, participants start walking, and the evaluator measured the time. The test was repeated twice and the lower of the two trials (in seconds) was recorded as the result.

4. Cardiorespiratory fitness outcomes

4.1 6-minute walk test

The 6-minute walk test, from the SFT battery, measured cardiorespiratory fitness. Participants were instructed to walk continuously for a duration of six minutes along a predetermined course, maintaining a comfortable yet brisk pace without resorting to running. The evaluator set up a rectangular course using four cones (18.3m x 4.57m) positioned at each corner. After explaining the test, the evaluator initiated the timing using a stopwatch, and the same evaluator concluded the test at the end of the six-minute interval. Subsequently, the distance (In meters) covered by the participant was the indicator of cardiorespiratory fitness. Higher scores reflect better cardiorespiratory fitness

4.2 2 km walking test

The 2 km walking test involved participants walking a designated path. The evaluator arranged four cones (28m x 15m) at each corner to create a rectangle, along with an additional cone 22 meters from the starting cone to mark the end of the 2 km distance. The evaluator explained the test, which required participants to complete 23 full laps plus an additional 22 meters to reach the end cone. Upon initiating the timer using a stopwatch, the evaluator monitored the participant's progress. The same evaluator concluded the test upon the participant's arrival at the 2 km mark. The time taken (in minutes) by the participant was reverted (multiplied by -1) and recorded. Higher scores reflect better cardiorespiratory fitness.

Supplementary material 6. Additional Tables and Figures

Table S10. Baseline descriptive characteristics of the AGUEDA sample.

	All	Exercise group	Control group
	Mean ± SD	Mean ± SD	Mean ± SD
Overall characteristics			
	90	46	44
Female, No. (%)	52 (57.8)	27 (58.7)	25 (56.8)
Age at baseline, yr	71.75 (3.96)	71.90 (4.21)	71.58 (3.73)
Education length, yr	11.54 (4.90)	11.17 (5.26)	11.93 (4.53)
12 yr of education or less, n (%)	57 (63.3)	32 (69.6)	25 (56.8)
Height, cm	160.44 (8.98)	160.17 (8.87)	160.72 (9.18)
Weight, kg	73.50 (12.86)	72.89 (13.41)	74.14 (12.39)
Body-mass index, kg/m²	28.50 (4.23)	28.35 (4.43)	28.67 (4.05)
More than 2 comorbidities , n (%)	37 (41.1)	15 (32.6)	22 (50.0)
ApoE ε4 carrier, n (%)	13 (14.8)	7 (15.6)	6 (14.0)
Amyloid positive, n (%)	19 (21.1)	8 (17.4)	11 (25.0)
Subjective cognitive decline, score	2.92 (1.97)	2.74 (1.99)	3.11 (1.94)
Cognitive status			
Telephone Interview of Cognitive Status, score	33.91 (2.72)	33.78 (2.65)	34.05 (2.81)
Mini-Mental State Examination , score	28.92 (1.06)	29.00 (0.97)	28.84 (1.16)
Montreal Cognitive Assessment, score	25.58 (2.14)	25.28 (2.21)	25.89 (2.05)
Cognitive function			
Executive function, z-score	-0.01 (1.00)	-0.03 (1.07)	0.01 (0.94)
Episodic memory, z-score	0.00 (1.01)	-0.01 (0.85)	0.01 (1.15)
Processing speed, z-score	-0.01 (1.00)	0.04 (0.85)	-0.06 (1.15)
Working memory, z-score	-0.02 (0.99)	0.01 (1.05)	-0.04 (0.94)
Attentional/inhibitory control, z-score	-0.01 (1.00)	0.04 (0.98)	-0.06 (1.03)
Visuospatial processing, z-score	-0.01 (1.00)	-0.09 (0.94)	0.08 (1.06)
Muscular strength by field tests			
Arm curl, n	16.00 (3.77)	15.80 (3.59)	16.20 (3.97)
30-seconds sit-to-stand test, n	13.98 (2.97)	13.91 (3.03)	14.05 (2.94)
5-times sit-to-stand test, sec	-9.20 (2.24)	-9.22 (2.32)	-9.19 (2.17)
Handgrip strength, kg	28.03 (9.67)	27.18 (8.80)	28.92 (10.53)

Table S10. Baseline descriptive characteristics of the AGUEDA sample.

	All	Exercise group	Control group
	Mean ± SD	Mean ± SD	Mean ± SD
Muscular strength by Isokinetic tests			
Elbow extension, N/m	89.90 (34.33)	87.25 (35.98)	92.67 (32.70)
Knee extension, N/m	95.89 (33.02)	93.08 (34.85)	98.83 (31.13)
Physical function			
2-Minute step, n	70.83 (19.18)	69.61 (21.72)	72.11 (16.26)
Up and go test*, sec	-5.81 (1.13)	-5.72 (1.01)	-5.92 (1.24)
Cardiorespiratory fitness			
6-Minute walk test, m	477.84 (73.33)	479.77 (73.58)	475.82 (73.87)
2 km walking test*, min	-23.29 (3.75)	-23.49 (4.13)	-23.08 (3.34)

Abbreviations: m, meters; n, number; N/m: Newton/meters, SD, standard deviation; sec, seconds; yr, years. *Reverted variable (*-1) for easy interpretation, higher score, the better performance.

Table S11. Intention-to-treat effects of the 24-weeks resistance exercise intervention on executive function for subgroups.

	Executive function		
	All-Pre	Exercise-Post	Control-Post
Sex			
Male	0.43 (0.18;0.68)	0.54 (0.08;1)	0.58 (0.09;1.07)
Female	-0.33 (-0.62;-0.04)	0.29 (0.01;0.56)	0.06 (-0.22;0.34)
Age (cut-off 72 years old)			
Youngers	0.22 (-0.02;0.45)	0.54 (0.18;0.9)	0.55 (0.18;0.93)
Olders	-0.29 (-0.65;0.07)	0.29 (-0.05;0.62)	-0.16 (-0.5;0.18)
Educational level (cut-off 12 years)			
High	0.47 (0.24;0.71)	0.67 (0.2;1.13)	0.82 (0.39;1.26)
Low	-0.29 (-0.57;-0.01)	0.27 (0;0.54)	-0.14 (-0.44;0.17)
APOE carrier			
Non-carrier	0.02 (-0.21;0.25)	0.5 (0.24;0.75)	0.31 (0.05;0.58)
Carrier	-0.2 (-0.85;0.46)	-0.11 (-1.14;0.91)	0.21 (-0.92;1.35)
Amyloid burden (cut-off 12 CL)			
Negative	-0.09 (-0.33;0.15)	0.36 (0.06;0.65)	0.27 (-0.05;0.6)
Positive	0.29 (-0.12;0.71)	0.51 (0.01;1.01)	0.28 (-0.18;0.74)
Executive function (median)			
High	0.7 (0.59;0.82)	0.98 (0.77;1.18)	1.01 (0.77;1.26)
Low	-0.73 (-1;-0.45)	-0.24 (-0.66;0.18)	-0.39 (-0.78;0)
Comorbidities (median)			
Low	-0.12 (-0.43;0.19)	0.36 (0;0.73)	0.22 (-0.18;0.62)
High	0.14 (-0.11;0.4)	0.36 (0;0.73)	0.29 (-0.07;0.64)
Subjective cognitive decline (median)			
Low	-0.1 (-0.4;0.19)	0.29 (-0.03;0.62)	0.44 (0.08;0.81)
High	0.15 (-0.11;0.41)	0.57 (0.16;0.98)	0.1 (-0.27;0.48)

Values indicate estimated marginal means at each time point (and 95% Confidence Intervals).

Table S12. Intention-to-treat effects of the 24-weeks resistance exercise intervention on cognitive domains for subgroups.

	Attentional/inhibitory control			Episodic memory			Processing speed			Visuospatial processing			Working memory		
	All-Pre	Exercise-Post	Control-Post	All-Pre	Exercise-Post	Control-Post	All-Pre	Exercise-Post	Control-Post	All-Pre	Exercise-Post	Control-Post	All-Pre	Exercise-Post	Control-Post
Sex															
Male	0.34 (0.05;0.64)	0.98 (0.55;1.41)	0.44 (-0.03;0.91)	0.09 (-0.25;0.43)	0.59 (0.02;1.17)	0.3 (-0.32;0.92)	0.4 (0.14;0.66)	0.93 (0.53;1.33)	0.68 (0.25;1.1)	0.15 (-0.21;0.52)	0.65 (0.16;1.14)	0.6 (0.05;1.15)	0.41 (0.1;0.71)	0.71 (0.25;1.17)	0.48 (-0.04;1)
Female	-0.27 (-0.55;0.01)	0.47 (0.16;0.77)	0.11 (-0.2;0.42)	-0.06 (-0.34;0.21)	0.3 (-0.07;0.68)	-0.04 (-0.41;0.34)	-0.31 (-0.6;-0.02)	0.05 (-0.28;0.37)	0.3 (-0.02;0.63)	-0.13 (-0.38;0.12)	-0.2 (-0.53;0.13)	-0.22 (-0.56;0.12)	-0.32 (-0.59;-0.06)	0.06 (-0.2;0.33)	0.09 (-0.18;0.37)
Age (cut-off 72 years old)															
Youngers	0.29 (0.05;0.53)	0.98 (0.64;1.32)	0.52 (0.16;0.88)	0.24 (0.02;0.46)	0.75 (0.32;1.18)	0.37 (-0.09;0.84)	0.15 (-0.08;0.39)	0.63 (0.31;0.95)	0.69 (0.36;1.02)	0.16 (-0.13;0.44)	0.34 (-0.04;0.72)	0.27 (-0.14;0.68)	0.11 (-0.16;0.38)	0.4 (0.1;0.7)	0.49 (0.17;0.81)
Olders	-0.39 (-0.73;-0.05)	0.32 (-0.04;0.67)	-0.09 (-0.45;0.27)	-0.3 (-0.67;0.08)	-0.05 (-0.51;0.41)	-0.22 (-0.68;0.25)	-0.21 (-0.58;0.16)	0.2 (-0.24;0.64)	0.14 (-0.31;0.59)	-0.22 (-0.53;0.1)	-0.07 (-0.56;0.43)	-0.12 (-0.63;0.39)	-0.17 (-0.5;0.16)	0.31 (-0.13;0.75)	-0.06 (-0.5;0.39)
Educational level (cut-off 12 years)															
High	0.37 (0.15;0.58)	1.04 (0.73;1.35)	0.73 (0.45;1.02)	0.45 (0.13;0.77)	1.01 (0.48;1.53)	0.81 (0.33;1.3)	0.51 (0.28;0.73)	1.06 (0.65;1.46)	1.04 (0.68;1.41)	0.46 (0.14;0.78)	0.48 (-0.13;1.09)	0.68 (0.14;1.21)	0.46 (0.19;0.73)	0.66 (0.18;1.13)	0.79 (0.36;1.21)
Low	-0.23 (-0.52;0.07)	0.51 (0.18;0.85)	-0.1 (-0.48;0.28)	-0.26 (-0.52;0)	0.1 (-0.27;0.47)	-0.4 (-0.83;0.04)	-0.31 (-0.59;-0.03)	0.1 (-0.22;0.42)	0.08 (-0.28;0.43)	-0.28 (-0.53;-0.03)	-0.01 (-0.32;0.3)	-0.32 (-0.69;0.05)	-0.29 (-0.56;-0.02)	0.21 (-0.06;0.47)	-0.18 (-0.5;0.13)
APOE carrier															
Non-carrier	0.01 (-0.23;0.24)	0.7 (0.4;1)	0.26 (-0.04;0.57)	0.04 (-0.2;0.28)	0.49 (0.13;0.85)	0.23 (-0.14;0.61)	0.01 (-0.22;0.24)	0.48 (0.17;0.78)	0.52 (0.21;0.83)	0.02 (-0.21;0.25)	0.17 (-0.14;0.48)	0.25 (-0.07;0.58)	0.06 (-0.15;0.28)	0.36 (0.1;0.62)	0.43 (0.15;0.7)
Carrier	-0.08 (-0.7;0.53)	0.52 (0.11;0.93)	0.16 (-0.28;0.6)	-0.24 (-0.76;0.29)	0.07 (-0.71;0.86)	-0.37 (-1.23;0.5)	-0.07 (-0.69;0.55)	0.21 (-0.4;0.81)	0.17 (-0.45;0.79)	-0.2 (-0.79;0.39)	0.38 (-0.52;1.28)	-0.7 (-1.74;0.34)	-0.54 (-1.29;0.2)	0.08 (-0.51;0.67)	-0.2 (-0.88;0.48)
Amyloid burden (cut-off 12 CL)															
Negative	-0.07 (-0.3;0.17)	0.64 (0.34;0.94)	0.15 (-0.18;0.47)	-0.04 (-0.29;0.21)	0.37 (-0.01;0.74)	0.08 (-0.32;0.48)	-0.02 (-0.26;0.21)	0.43 (0.13;0.74)	0.43 (0.1;0.76)	0.03 (-0.21;0.28)	0.16 (-0.17;0.48)	0.3 (-0.07;0.66)	-0.08 (-0.33;0.17)	0.35 (0.08;0.62)	0.35 (0.05;0.65)
Positive	0.2 (-0.28;0.68)	0.86 (0.47;1.24)	0.59 (0.22;0.96)	0.15 (-0.26;0.55)	0.75 (-0.27;1.76)	0.17 (-0.62;0.97)	0.05 (-0.43;0.53)	0.38 (-0.15;0.9)	0.57 (0.07;1.08)	-0.16 (-0.6;0.28)	0.34 (-0.46;1.15)	-0.55 (-1.25;0.15)	0.23 (-0.1;0.56)	0.24 (-0.47;0.94)	0.04 (-0.54;0.63)
Comorbidities (median)															
Low	-0.02 (-0.31;0.26)	0.76 (0.42;1.1)	0.13 (-0.24;0.49)	-0.08 (-0.36;0.2)	0.51 (0.09;0.93)	-0.01 (-0.47;0.45)	-0.08 (-0.4;0.25)	0.37 (0;0.74)	0.41 (0.02;0.8)	0.02 (-0.24;0.29)	0.21 (-0.17;0.58)	0.03 (-0.39;0.46)	-0.07 (-0.37;0.23)	0.33 (0.02;0.63)	0.25 (-0.1;0.59)
High	0.01 (-0.31;0.33)	0.53 (0.15;0.9)	0.43 (0.07;0.79)	0.12 (-0.21;0.45)	0.14 (-0.36;0.63)	0.33 (-0.16;0.82)	0.09 (-0.14;0.32)	0.5 (0.11;0.9)	0.54 (0.17;0.92)	-0.06 (-0.41;0.3)	0.15 (-0.38;0.67)	0.15 (-0.35;0.64)	0.06 (-0.23;0.35)	0.36 (-0.11;0.84)	0.28 (-0.17;0.73)
Subjective cognitive decline (median)															
Low	-0.08 (-0.37;0.21)	0.64 (0.35;0.92)	0.43 (0.12;0.74)	-0.14 (-0.42;0.15)	0.37 (-0.01;0.76)	0.26 (-0.18;0.69)	-0.03 (-0.31;0.26)	0.37 (0.06;0.68)	0.63 (0.29;0.97)	-0.15 (-0.42;0.12)	0.15 (-0.19;0.49)	0.32 (-0.09;0.72)	-0.09 (-0.34;0.17)	0.31 (0.04;0.57)	0.51 (0.21;0.81)
High	0.11 (-0.19;0.4)	0.77 (0.28;1.26)	0.09 (-0.36;0.53)	0.24 (-0.07;0.54)	0.42 (-0.15;1)	0.08 (-0.46;0.62)	0.03 (-0.29;0.34)	0.53 (0.06;1.01)	0.28 (-0.17;0.73)	0.23 (-0.1;0.57)	0.15 (-0.41;0.72)	-0.07 (-0.58;0.43)	0.1 (-0.28;0.49)	0.43 (-0.07;0.94)	-0.03 (-0.48;0.42)
Baseline cognitive domain															
High	0.79 (0.66;0.91)	1.12 (0.79;1.45)	0.93 (0.59;1.26)	0.76 (0.59;0.93)	1.14 (0.74;1.55)	0.89 (0.49;1.29)	0.72 (0.59;0.85)	1.07 (0.83;1.32)	1.28 (1.02;1.55)	0.82 (0.68;0.96)	0.72 (0.26;1.18)	0.73 (0.26;1.2)	0.79 (0.66;0.93)	0.78 (0.41;1.15)	0.74 (0.36;1.13)
Low	-0.81 (-1.03;-0.58)	0.2 (-0.12;0.51)	-0.38 (-0.72;-0.05)	-0.76 (-0.98;-0.54)	-0.28 (-0.69;0.13)	-0.72 (-1.18;-0.26)	-0.74 (-1;-0.48)	-0.22 (-0.58;0.14)	-0.33 (-0.69;0.03)	-0.8 (-1;-0.6)	-0.28 (-0.58;0.03)	-0.6 (-0.95;-0.25)	-0.83 (-1.03;-0.62)	-0.11 (-0.39;0.18)	-0.24 (-0.56;0.07)

Values indicate estimated marginal means at each time point (and 95% Confidence Intervals).

Table S13. Intention-to-treat effects of the 24-weeks resistance exercise intervention on physical parameters (raw data).

	Change from Baseline to 24-weeks	
	Exercise	Control
Muscular strength by field tests		
Arm curl, n	18.49 (17.5;19.47)	18.19 (17.15;19.22)
30-seconds sit-to-stand test, n	16.65 (15.59;17.7)	15.49 (14.38;16.59)
5-times sit-to-stand test, sec	-8.14 (-8.75;-7.54)	-8.6 (-9.23;-7.96)
Handgrip strength, kg	27.88 (25.86;29.9)	27.24 (25.2;29.28)
Muscular strength by Isokinetic tests		
Elbow extension, N/m	96.06 (88.56;103.57)	92.42 (84.42;100.42)
Knee extension, N/m	92.69 (86.09;99.29)	84.31 (77.41;91.21)
Physical function		
Two-Minute step, n	88.1 (81.24;94.95)	87.01 (79.67;94.35)
Up and go test, sec	-5.39 (-5.7;-5.09)	-5.45 (-5.77;-5.13)
Cardiorespiratory Fitness		
6-Minute walk test, m	508.1 (489.4;526.79)	498.02 (478.61;517.43)
2 km walking test, min	-22.48 (-23.39;-21.57)	-23.34 (-24.29;-22.39)

Abbreviations: m, meters; n, number; N/m: Newton/meters, SD, standard deviation; sec, seconds; yr, years. Values indicated estimated marginal means (and 95% Confidence Intervals) after 24 weeks of exercise intervention.

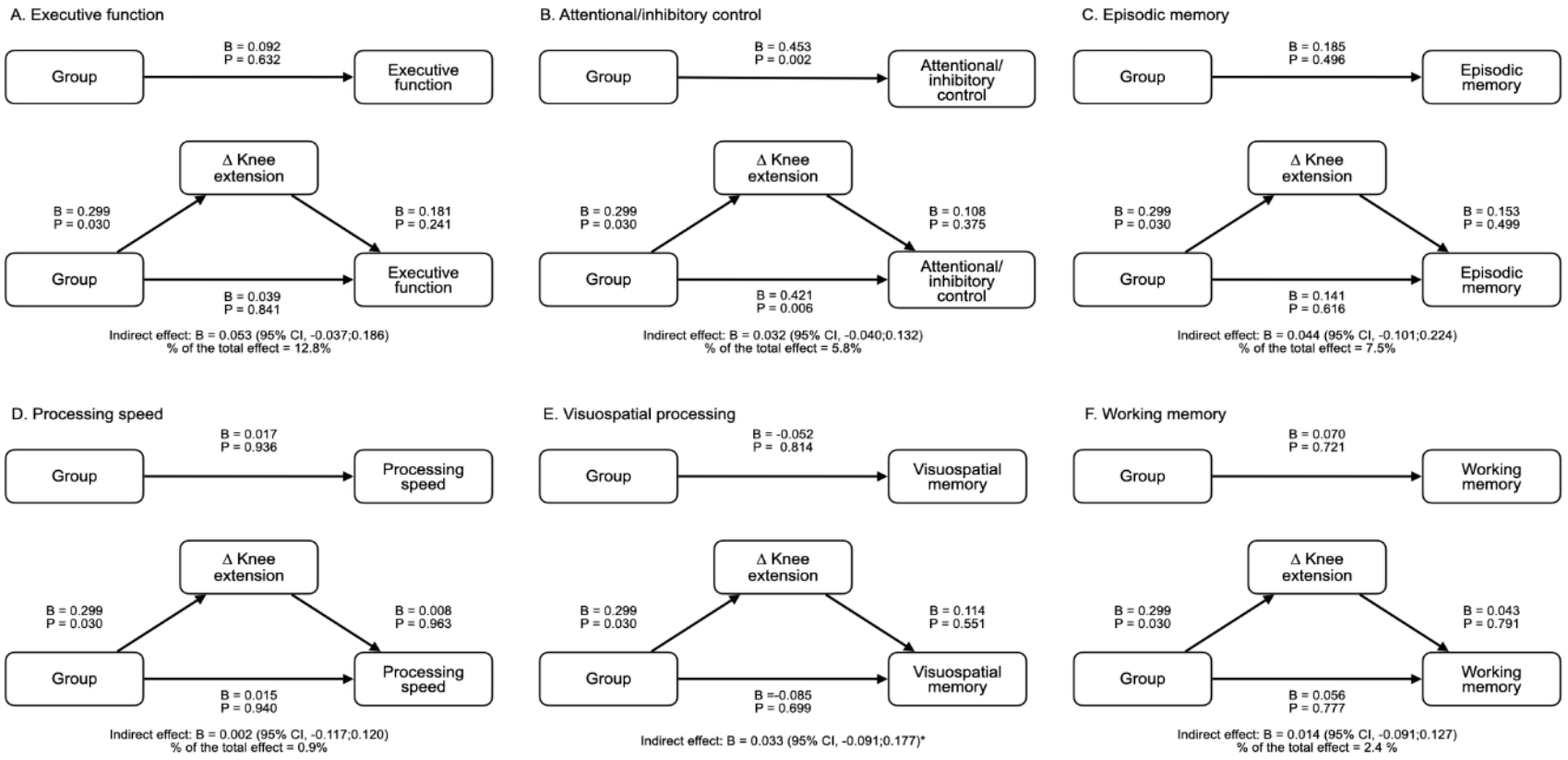


Figure S3. Mediation models of the Intervention effects knee extension on executive function and cognitive domains. *Negative % of mediation. △Knee extension represents the variable of change.

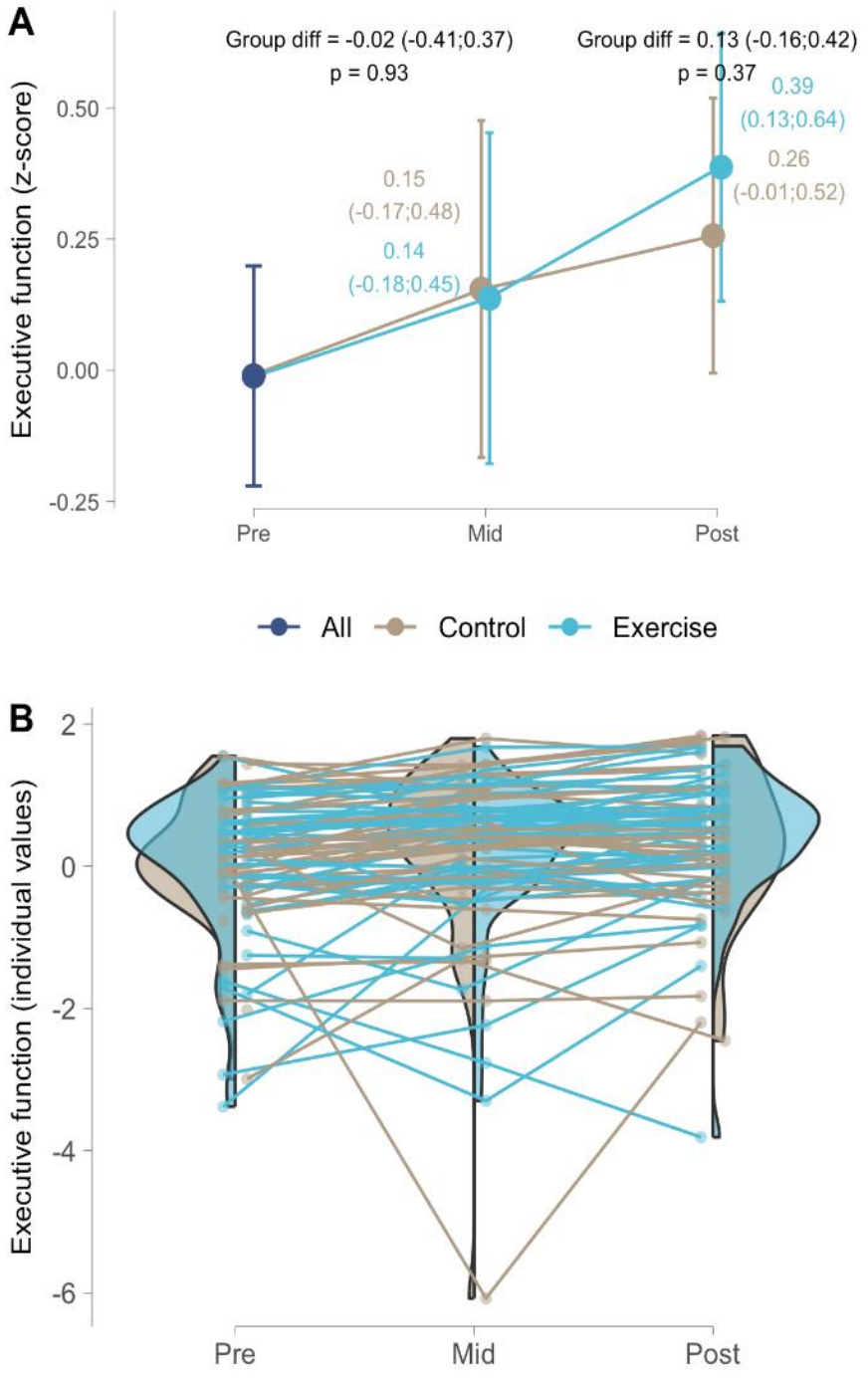


Figure S4. Intention-to-treat effects at 12 and 24 weeks of the resistance exercise intervention on executive function showing mean values (A) and individual values (B). A. Dots indicate estimated marginal means at each time point (and 95% Confidence Intervals). B. Dots indicate participant individual raw values.

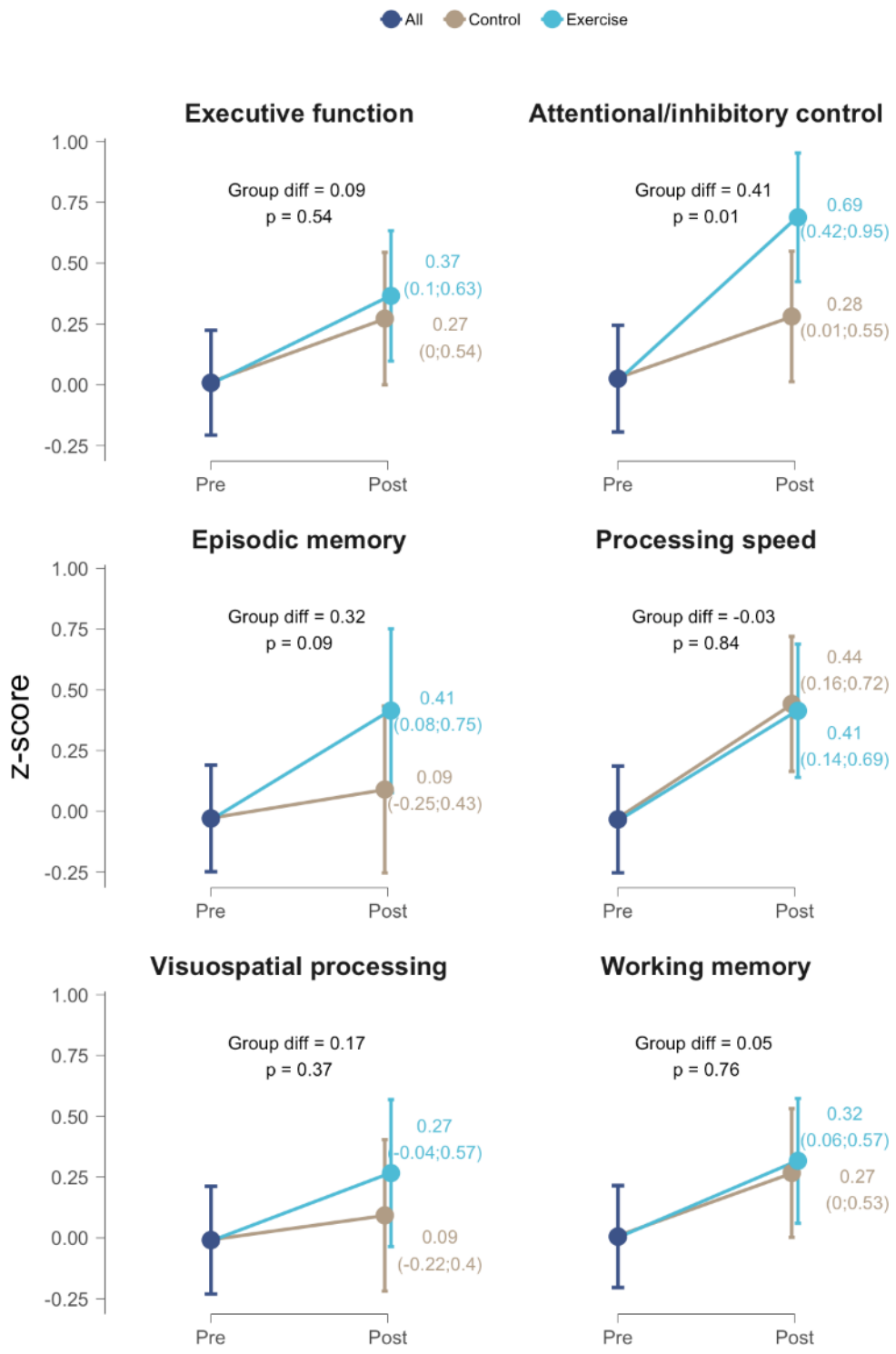


Figure S5. Per-protocol effects of the 24-week resistance exercise program on executive function and other cognitive domains. Dots indicate estimated marginal means at each time point (and 95% Confidence Intervals).

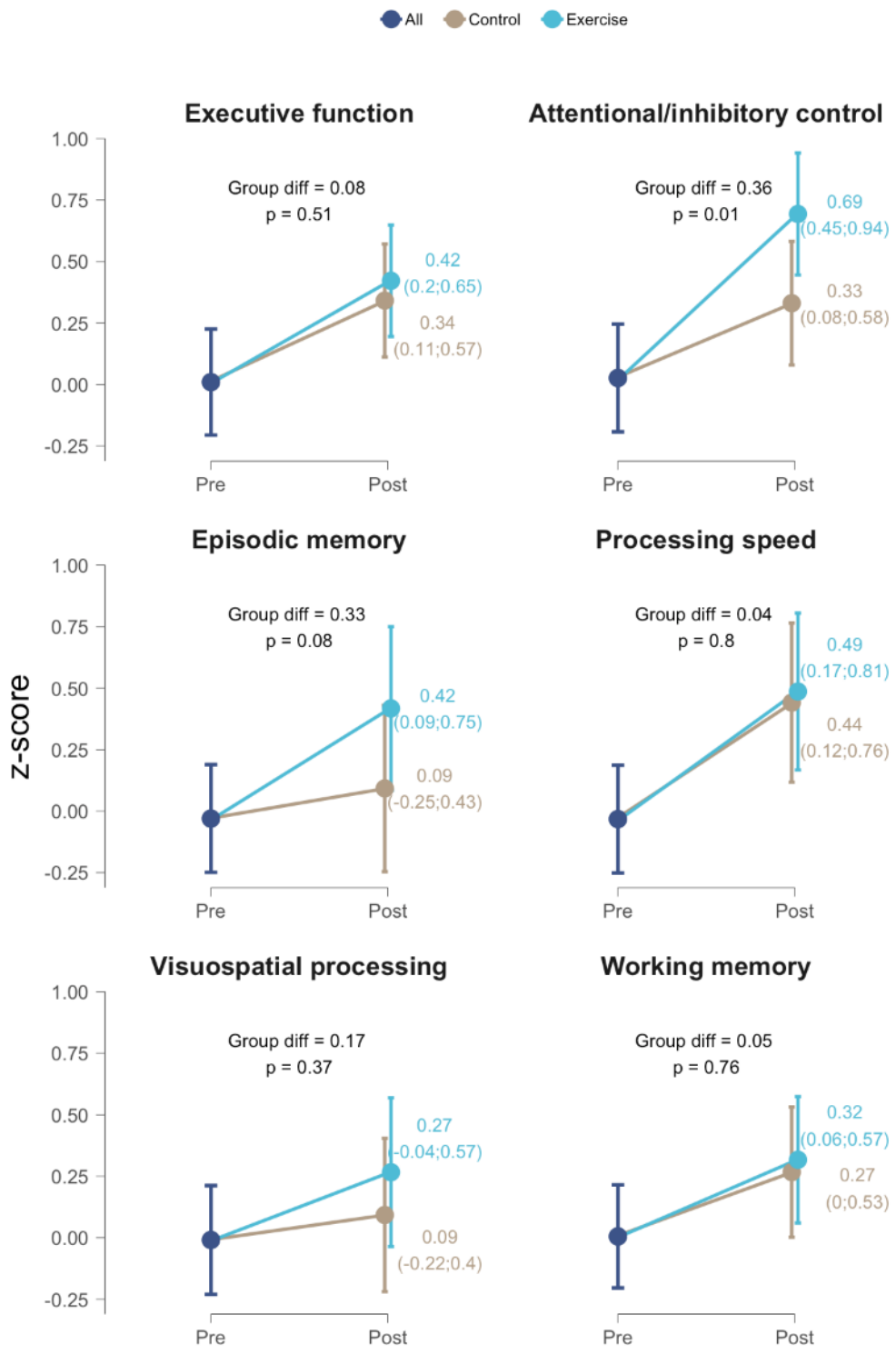


Figure S6. Per-protocol effects of the 24-week resistance exercise intervention on executive function and cognitive domains. Dots indicate estimated marginal means at each time point (and 95% Confidence Intervals). Winsorized values >4S

Table S14. Adverse events during 24-week period of exercise intervention.

Adverse event #	Severity adverse event*	Group	Reason	Relationship with the Agueda Exercise intervention	Excluded from**	Action	Week of adverse event
1	Mild	Exercise group	Illness	No	Biological variables	Recommendations to visit the doctor	Week 17
2	Mild	Exercise group	Knee pain	No	Lower body physical variables	Modifications and adaptations of the exercises	Week 22
3	Moderate	Exercise group	Shoulder injury	No	Upper body physical variables	Modifications and adaptations of the exercises	Week 13
4	Moderate	Exercise group	Shoulder injury	No	Upper body physical variables	Rehabilitation modifications and adaptations of the exercises	Week 2
5	Moderate	Exercise group	Back pain	No	Upper and lower body physical variables	Modifications and adaptations of the exercises	Week 21
6	Severe	Exercise group	Shoulder surgery	No	Upper body physical variables	Rehabilitation Modifications and adaptations of the exercises	Week 5
7	Severe	Control group	Stage I of Cancer	No	All the variables	-	Week 12
8	Severe	Control group	Stroke	No	All the variables	-	Week 18

*Mild, moderate or severe. **Those participants that had the adverse event closer to the 12 week or 24-week evaluations were excluded in sensitivity analyses

Supplementary material 7. Additional references

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Active Gains in brain Using Exercise During Aging (AGUEDA): protocol for a randomized controlled trial

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Alzheimer's disease is currently the leading cause of dementia and one of the most expensive, lethal and severe diseases worldwide. Age-related decline in executive function is widespread and plays a key role in subsequent dementia risk. Physical exercise has been proposed as one of the leading non-pharmaceutical approaches to improve executive function and ameliorate cognitive decline. This single-site, two-arm, single-blinded, randomized controlled trial (RCT) will include 90 cognitively normal older adults, aged 65–80 years old. Participants will be randomized to a 24-week resistance exercise program (3 sessions/week, 60 min/session, n = 45), or a wait-list control group (n = 45) which will be asked to maintain their usual lifestyle. All study outcomes will be assessed at baseline and at 24-weeks after the exercise program, with a subset of selected outcomes assessed at 12-weeks. The primary outcome will be indicated by the change in an executive function composite score assessed with a comprehensive neuropsychological battery and the National Institutes of Health Toolbox Cognition Battery. Secondary outcomes will include changes in brain structure and function and amyloid deposition, other cognitive outcomes, and changes in molecular biomarkers assessed in blood, saliva, and fecal samples,

physical function, muscular strength, body composition, mental health, and psychosocial parameters. We expect that the resistance exercise program will have positive effects on executive function and related brain structure and function, and will help to understand the molecular, structural, functional, and psychosocial mechanisms involved.

KEYWORDS

exercise, executive function, Alzheimer's disease, amyloid beta, brain

1. Introduction

Dementia is generally characterized by a progressive degeneration of the brain, followed by deterioration of both cognitive abilities and the capacity to perform activities of daily living (Guha, 2014; Petermann-Rocha et al., 2020). Alzheimer's disease (AD) is currently the leading cause of dementia, and one of the most expensive, lethal and serious diseases (Scheltens et al., 2021). Interestingly, the development of the neuropathological features of AD begins more than 20 years before the onset of related clinical symptoms (Bateman et al., 2012). One of the main markers of neuropathology that can appear before the onset of clinical symptoms is the pathological accumulation of brain amyloid beta (Aβ), which is present in more than 20% of cognitively normal older adults (Jansen et al., 2022). Evidence of pathological levels of this peptide may precede changes in brain structure and function as well as cognitive decline (Erickson et al., 2019); particularly, executive function, which refers to high-level, goal-directed cognitive processes involved in activities of daily living, together with memory performance, are among the earliest compromised cognitive domains (Kirova et al., 2015; Tideman et al., 2022). However, while memory decline is one of the most typical changes in AD, impaired executive function may precede (Harrington et al., 2013; Ho et al., 2018) and be predictive of additional cognitive decline (Kirova et al., 2015; Kim et al., 2020; Shepherd et al., 2021), development of mild cognitive impairment (MCI) and AD (Gibbons et al., 2012), especially at older ages (Mez et al., 2013; Petermann-Rocha et al., 2020), and ultimately being a potential target of lifestyle interventions due to their potentially non-specific cognitive benefits (Erickson et al., 2022).

Dementia is not a natural or inevitable consequence of aging (Verissimo et al., 2022). The risk of experiencing cognitive decline may be ameliorated by a physically active lifestyle (Livingston et al., 2020; Feter et al., 2021; Wu et al., 2022). Indeed, up to 20% of the risk of developing dementia in the world's population can be attributed to physical inactivity, and it is estimated that around 10 million new cases might be avoided or delayed worldwide each year if older adults meet recommendations for physical activity (World Health Organization [WHO], 2015). Thus, physical exercise (i.e., structured and programmed subset of physical activity) emerges as a potential non-pharmaceutical preventive and cost-efficient treatment for age-related cognitive decline, having an influence at molecular, structural and functional, and behavioral levels (Chan et al., 2021).

Several systematic reviews and meta-analyses have described the beneficial effect of exercise interventions on cognitive domains in older adults, including executive function (Hindin and Zelinski, 2012; Guiney and Machado, 2013; Miller and Taylor-Piliae, 2014; Ohman et al., 2014; Wayne et al., 2014; Lu et al., 2016; Gomes-Osman et al., 2018; Chen et al., 2020), memory (Lu et al., 2016), language (Lu et al., 2016), and general cognition (Miller and Taylor-Piliae, 2014; Lu et al., 2016; Barha et al., 2017; Gomes-Osman et al., 2018; Northey et al., 2018); however, there are also studies showing a lack of clear benefits (Lautenschlager et al., 2008; van Uffelen et al., 2008; Snowden et al., 2011; Tsai et al., 2015; Hong et al., 2018; Lamb et al., 2018; Vidoni et al., 2021). Previously conducted randomized controlled trials (RCTs) have primarily focused on the effects of aerobic exercise (i.e., walking) on cognition in cognitively normal older adults (Erickson et al., 2011, 2019; Stillman et al., 2020; Mendez Colmenares et al., 2021), and in adults at early stages of AD (Zheng et al., 2016; Stillman et al., 2020). Resistance exercise (i.e., muscular strength training) is another type of exercise with the potential for improving cognition (Yoon et al., 2018). Although this type of exercise is included in the physical activity guidelines (Bull et al., 2020), the effects of resistance exercise on cognition have been less frequently studied compared to aerobic exercise (Yoon et al., 2018). Interestingly, recent reviews indicate that resistance exercise interventions (over other types of exercise such as walking or yoga, among others) might have a substantial effect on cognition (Li et al., 2018; Northey et al., 2018; Gallardo-Gomez et al., 2022), including at relatively low doses (Gallardo-Gomez et al., 2022). In general, previous studies have found a positive effect of resistance exercise on executive function (Liu-Ambrose et al., 2008; Brown et al., 2009; Coetsee and Terblanche, 2017) and general cognition (Li et al., 2018) in older adults, while there are less clear results on prevention of cognitive decline (Colcombe and Kramer, 2003). However, these interventions differed from each other, in terms of duration (6 weeks to 1 year) (Liu-Ambrose et al., 2008, 2010; Brown et al., 2009; Mavros et al., 2017), type of resistance exercise (i.e., free weight exercises, elastic band exercises, dumbbells/barbells exercises), and frequency (i.e., 1–3 sessions/week), making it difficult to determine the specific dose and type of resistance exercise needed to improve cognition (Li et al., 2018). Thus, these somewhat inconsistent findings call for more well-designed RCTs with the capacity to clarify the effects of resistance exercise on cognition in older adults.

Potential mechanisms underlying the beneficial effects of exercise on cognition have been recently investigated. A recent conceptual review of this literature suggests three evidence-based levels of mechanisms demonstrated by exercise-based RCTs: (i) *molecular level*, such as increments in brain derived neurotrophic factor (BDNF) and insulin-like growth factor 1

(IGF-1); (ii) *brain structure and function level,* such as preserving hippocampal volume, changes in cortical and white matter morphology and changes in functional connectivity; and (iii) *behavioral level,* such as improvements in mood and sleep (Stillman et al., 2020). Not surprisingly, evidence and discussion around most of these mechanisms are demonstrated by aerobic exercise RCTs. However, different types of exercise (i.e., aerobic or resistance exercise) may involve distinct pathways at each mechanistic level (Vilela et al., 2017). For example, some animal studies suggest that the improvements in cognition derived from resistance exercise may be related to other central and peripheral inflammatory changes (Vilela et al., 2017), or metabolic profile (Serra et al., 2021), and a few human studies highlight the release of neurochemical markers such as BDNF, IGF-1, and homocysteine (Liu-Ambrose and Donaldson, 2009; Herold et al., 2019). On the other hand, effects of other novel brain-related markers of aerobic exercise such as epigenetic factors (Liang et al., 2021; Urdinguio et al., 2021), Cathepsin B (Pedersen, 2019), vascular endothelium growth factor (VEGF), glycosylphosphatidylinositol-specific phospholipase D1 (GPLD1) (Horowitz et al., 2020), clusterin (Horowitz et al., 2020) and central markers such as cerebral blood flow, are underexplored related to resistance exercise interventions in humans. Thus, examining the mechanisms underlying the effects of resistance exercise on cognition is needed, particularly in cognitively normal older adults. To that end, well-designed resistance exercise RCTs to test its effects on cognition, and the potential mechanisms involved should be implemented. In addition, those studies should adequately report the exercise characteristics (i.e., type, frequency, and volume) to establish coherent and robust evidenced-based exercise recommendations (Huang et al., 2021).

Here, we propose the Active Gains in brain Using Exercise During Aging (AGUEDA) trial to shed light on the research gaps related to resistance exercise. This paper describes the rationale and methodology of the AGUEDA RCT. The primary aim of the study is to investigate the effects of a 24-week resistance exercise program on executive function in cognitively normal older adults. The secondary aims are (i) to examine the effects of resistance exercise on brain structural and functional markers (e.g., gray and white matter measures, functional connectivity or cerebral blood flow), brain Aβ deposition, peripheral molecular markers (e.g., BDNF, IGF-1, and Aβ), and other cognitive outcomes (e.g., memory and language), and (ii) to investigate mediators and moderators of the potential exercise-derived improvements observed in executive function and brain markers. The overall hypothesis is that a 24-week resistance exercise program will improve executive function and related brain health metrics in cognitively normal older adults. In addition, we hypothesized that mechanisms involved in exercise-induced enhancements to executive function are present at molecular, brain structure and function, and behavioral levels.

2. Methods and analysis

2.1. Design and ethics

The AGUEDA trial is a single-center, two-arm, single-blind RCT in which 90 cognitively normal older adults, aged 65–80 years old, will be randomized into a resistance exercise group (n = 45 or a wait-list control group (n = 45). A visual representation of the participant flow during the study can be seen in **Figure 1**. The study will take place in Granada (Spain) at the Sport and Health University Research Institute (iMUDS) and Mind, Brain and Behavior Research Centre (CIMCYC) from the University of Granada, and the Virgen de las Nieves University Hospital. The trial protocol is in accordance with the principles of the Declaration of Helsinki and has been approved by the Research Ethics Board of the Andalusian Health Service (CEIM/CEI Provincial de Granada; #2317-N-19) on May 25th, 2020. The trial is registered on Clinicaltrials.gov (ClinicalTrials.gov Identifier: NCT05186090; Submission date: December 22, 2021). All participants will provide written informed consent once all study details have been explained. A model consent form is publicly available in github.[1] The study has been designed following the Standard Protocol Items for Randomized Interventional Trials (SPIRIT) (Chan et al., 2013), and the SPIRIT checklist filled for the present study can be found in **Supplementary Table 1**. In the event that any important protocol modifications are adopted, the study team will communicate them to the trial registry and Research Ethics Board according to the SPIRIT recommendations, and posted in github (see text footnote 1). In addition, all participants of the AGUEDA trial are covered by a social responsibility insurance policy provided by the University of Granada, in case ancillary, post-trial care or compensation are needed. Participants will not receive any monetary incentive for participating in the study.

2.2. Recruitment

We began recruitment of community-dwelling older adults in Granada (Spain) city and surrounding areas in March 2021. Strategies to achieve the targeted sample size mainly include mass mailings and social media advertisements, word of mouth, with augmentation of the recruitment strategy by using advertisements in newspapers (TV, radio, and internet). We plan to enroll 90 cognitively normal older adults between the ages of 65 and 80 years, that meet the inclusion and exclusion criteria presented in **Table 1**. Recruitment, enrollment, and randomization occurs on a rolling basis.

2.3. Screening

2.3.1. Telephone screening

Following recruitment strategies and after the participant's first contact for general information, an initial telephone interview is performed with those individuals who are potentially interested in participating. Questions about structured exercise and physical activity levels, medical history, Magnetic Resonance Imaging (MRI) compatibility and demographic characteristics are provided by a trained evaluator in the interview. In addition, the Spanish version of the modified Telephone Interview for Cognitive Status (STICS-m) is administered (Munoz-Garcia et al., 2019), and only those that score ≥26 points are eligible to participate. After the

1 https://github.com/aguedaprojectugr/Protocolos_AGUEDA

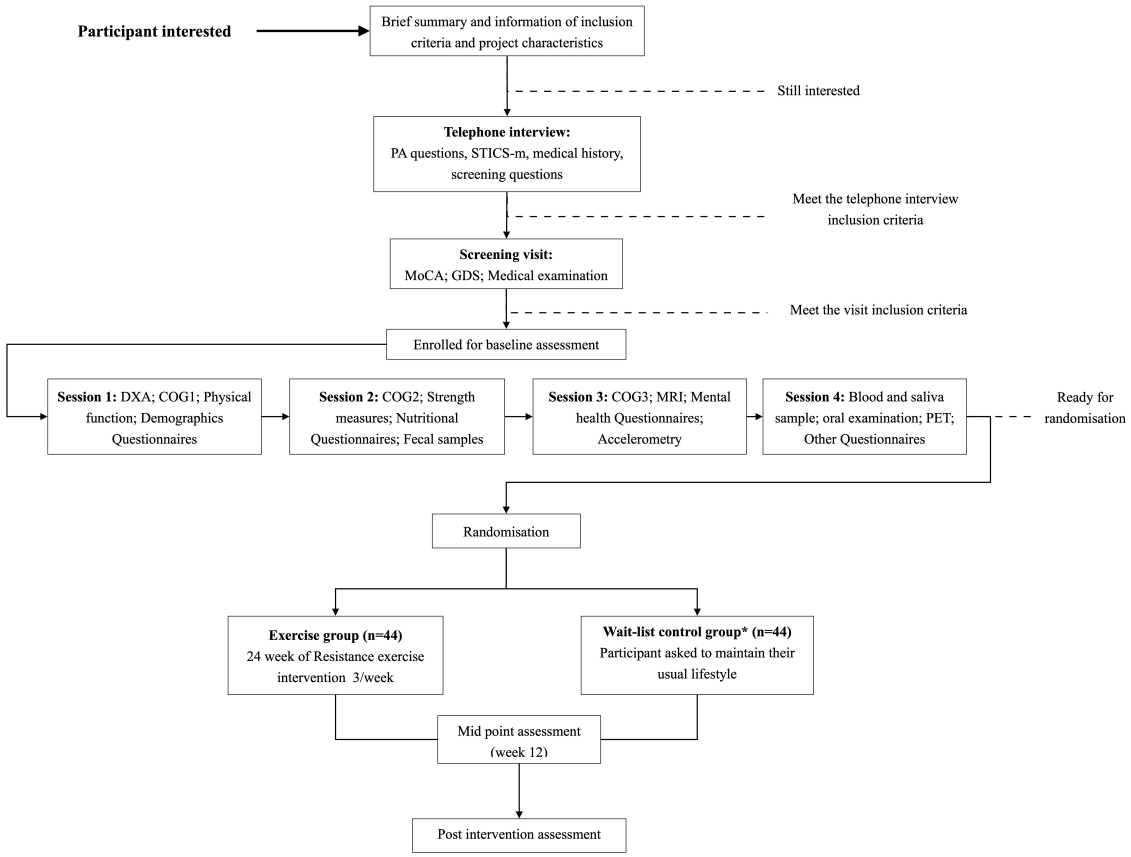


FIGURE 1
Visual representation of the participant flow in the AGUEDA trial. PA, physical activity; STICS-m, Modified Spanish Telephone Interview of Cognitive Status; MoCA, Montreal Cognitive Assessment; GDS, Geriatric Depression Scale; DXA, dual energy X-ray absorptiometry; MMSE, Mini-Mental State Examination; MRI, Magnetic Resonance Imaging; PET, positron emission tomography. *Wait-list control group is asked to maintain its lifestyles during the 24-week and start the exercise program after post-intervention assessment. #Post-intervention assessment includes the same outcomes that baseline assessment.

interview, an initial check of inclusion/exclusion criteria met by the participants in accordance to their medical history, physical activity levels, MRI safety, and cognitive status is performed before the screening visit.

2.3.2. On-site screening

Before the enrollment of participants in baseline assessments, an on-site screening will be performed. To determinate whether participants meet inclusion/exclusion criteria, this screening includes the Montreal Cognitive Assessment (MoCA) test, the Mini-Mental State Examination (MMSE, performed in a different visit than the MoCA), the Geriatric Depression Scale (GDS) (Izal and Montorio, 1993), and a medical examination conducted by a physician to determinate safety participate in resistance exercise program (**Table 1**). The MoCA and MMSE both have a maximum of 30-points and are used to identify cognitively normal older adults. We used standardized cut-offs according to age (<71 years, ≥24/30, 71–75 ≥ 22/30, >75, 21/30) and adjusted by years of education for a Spanish population for the MoCA (Ojeda et al., 2016), and ≥25/30 for MMSE. The GDS-30 is used to determine whether the participant meets the exclusion criteria for depressive symptoms. Any participant with a GDS ≥ 15 was excluded from further participation (Fernández-San Martín et al., 2002).

2.4. Outcome measurements

A summary of the primary, secondary, and additional outcomes and their time-points of collection is shown in **Table 2**. All outcome-related measures and analyses are performed by staff who were blinded to the intervention assignment. In brief, the primary outcome of the AGUEDA trial is an executive function composite score that includes several tests assessing domains of executive function such as cognitive flexibility, working memory and inhibitory control (Diamond, 2013). Secondary outcomes include additional cognitive measures (e.g., memory, language, and intelligence), as well as brain structure and function, brain Aβ load and selective peripheral biomarkers. Other health-related outcomes will be evaluated to test for mediation and moderation of biological, physiological, mental health and physical function variables on the potential exercise-derived improvements observed in executive function and brain markers. All protocols can be found in its original language (Spanish) in github (see text footnote 1).

2.4.1. Executive function and other cognitive outcomes

Executive function was selected as the primary outcome based on its importance for general cognitive decline and risk for

TABLE 1 Active Gains in brain Using Exercise During Aging (AGUEDA) inclusion and exclusion criteria for selecting participants.

Inclusion criteria	Exclusion criteria
- Men and women 65–80 years old.	- Ambulatory with pain or regular use of an assisted walking device.
- Able to speak and read fluent Spanish.	- Medical contraindication for inclusion in an exercise program.
- Living in community settings during the study.	- Neurological condition (multiple sclerosis, Parkinson's disease, dementia) or brain injury (traumatic or stroke).
- Reliable means of transportation.	- Current diagnosis and treatment of a DSM V axis I or II disorder including major depression, and seeing a psychologist, therapist, or psychiatrist in the last year.
- Being physically inactive: (i) not participating in any resistance exercise programs in the last 6 months, and (ii) accumulating less than 600 METs/week of moderate-vigorous physical activity.	- History of major psychiatric illness including schizophrenia, general anxiety disorder, or depression (GDS-30 $\geq$ 15).
- Classified as cognitively healthy according to STICS-m, MoCA, and MMSE.	- Current treatment for congestive heart failure, angina, uncontrolled arrhythmia, deep venous thrombosis or another cardiovascular event.
	- Myocardial infarction, coronary artery bypass grafting, angioplasty or other cardiac condition in the last year.
	- Current or previous treatment for any type of cancer.
	- Type I diabetes or uncontrolled Type II diabetes defined as insulin dependent.
	- Current treatment for alcohol or substance abuse.
	- Presence of metal implants (e.g., pacemaker, stents, and joint replacement) that would be MRI ineligible.
	- Claustrophobia.
	- Color blindness.
	- Diagnosis of COVID-19 with hospitalization in intensive care unit.
	- Any other consideration that interferes with the study aims and could be a risk to the participant, at the discretion of the researcher

DSM, Diagnostic and Statistical Manual of Mental Disorders; GDS, Geriatric Depression Scale; METs, Metabolic Equivalents; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; STICS-m, Modified Spanish Telephone Interview of Cognitive Status; MRI, Magnetic Resonance Imaging; COVID-19, coronavirus disease-19.

dementia in older adults (Vaughan and Giovanello, 2010). Briefly, the executive function composite score will be computed using scores from both paper-and-pencil tests (Trail making test A and B and Digit symbols substitution test) and computerized-based National Institutes of Health (NIH) Toolbox tests (Dimensional Card Sort Task, List Sort Working Memory, Picture Sequence and Flanker test), and programmed tests in E-prime software (task-switching, spatial working memory and Stroop test). Individual task scores will be converted to standardized z-scores by subtracting individual raw values minus the baseline group mean, and dividing by the baseline group standard deviation. First, we will verify whether the z-score of each task falls under the proposed executive function domains (i.e., cognitive flexibility, working memory and inhibitory control) using confirmatory factor analysis. Then, the z-score of each task will be equally weighted and pooled into its domain, and then averaged them into an executive function composite score. Additional cognitive outcomes will be assessed such as general cognition, language, verbal memory, visuospatial memory, and crystallized and fluid intelligence. **Table 3** shows the cognitive tests and their time point of collection, and details the tests incorporated into the executive function composite score. To ensure accurate assessment, all paper-and-pencil tests will be scored by two different trained staff members and in cases of disagreement, a solution will be resolved by consensus. In addition, the order of the tests (paper-and-pencil and computerized-based tests) will be randomized considering the assessed cognitive domain to minimize fatigue from affecting the results.

2.4.2. Brain structure and function

Brain structure and function will be assessed by MRI at the CIMCYC from the University of Granada. Participants will wear comfortable clothes and sign a specific MRI informed consent. As per the informed consent, participants will have to be compatible with MRI, i.e., not have metallic implants that could be dangerous in the MRI environment and report not being claustrophobic. Standard sequences will be collected for a total acquisition time of approximately 60 min. **Table 4** shows the image sequences that will be performed for each participant in acquisition order. A Siemens Magnetom PRISMA Fit 3T scanner with a 64-channel head coil will be used and all brain sequences will have the same slice selection. This slice selection from the field map will be copied to each sequence except for the focal hippocampal sequence, which has its own instructions for determining coverage of the anatomy. Notably, a radiologist will review structural sequences to check for any incidental findings. In cases of incidental findings, the radiologist will contact the participant for further examination. To ensure a minimal loss of data, we will perform a visual inspection and a quality control check of the images on a rolling basis. In addition, custom quality control will be performed including sequence accuracy, conversion to brain imaging data structure (BIDS) format, a check of sequence parameters by examining header information, and quantitative control using MRQC software (Esteban et al., 2017). Briefly, data collected will consist of images providing information about brain volume, surface, shapes and thickness, white matter microstructure, resting state functional MRI and task-evoked fMRI, and cerebral blood flow.

2.4.3. Brain amyloid beta deposition

The brain Aβ deposition will be assessed before and after intervention using a PET scan Biograph-Vision 600 Edge Positron

TABLE 2 Assessments' organization and distribution across time-points.

	Telephone-screening	On-site screening	Baseline	Mid-point	Post-intervention
Demographics	x	x			
Medical history	x	x			
Modified Spanish Telephone Interview of Cognitive Status (STICS-m)	x				
Physical activity level (self-reported)	x				
Medications		x			
Family history		x			
Montreal cognitive assessment (MoCA)		x			x
Mini-Mental State Examination (MMSE)		x			x
Geriatric Depression Scale (GDS)		x		x	x
Medical examination		x			
Primary outcome					
Executive function tests			x	x	x
Secondary outcomes					
Positron emission tomography (PET)			x		x
Magnetic Resonance Imaging (MRI)			x		x
Other cognitive tests			x		x
Other outcomes					
Muscular strength			x	x	x
Physical function			x	x	x
Physical activity monitoring			x	x	
Body composition and blood pressure			x		x
Biological samples (blood, fecal and saliva)			x		x
Mental health and other questionnaires			x	x	x

emission tomography/Computed tomography (PET/CT) digital scanner (Siemens, Erlangen, Germany) at the Virgen de las Nieves University Hospital in Granada. The accumulation of Aβ is a hallmark of AD and may begin up to 20 years before the onset of dementia (Bateman et al., 2012). For the AGUEDA trial, the tracer used will be [^{18}F]Florbetaben (Neuraceq; Piramal Pharma). [^{18}F]Florbetaben will be injected intravenously into each participant by a professional nurse blinded to intervention in accordance with the applicable regulatory guidelines. PET images will be acquired from participants 90 to 110 min after intravenous injection of 300 megabecquerel (MBq) ± 20% (Villemagne et al., 2011; Sabri et al., 2015a) according to a standardized acquisition and image-processing protocol (Barthel et al., 2011; Sabri et al., 2015b; Minoshima et al., 2016). Before the acquisition of PET images, a low-frequency CT scan will be performed and used for attenuation correction of PET images (5 min). PET/CT imaging (see **Table 4** for parameter's details) will be performed by a blinded specialist physician for all study participants. A visual quality control after each scan will be performed to ensure quality of data. In addition, a neurologist will visually review the PET images to check for any suspected findings, and only in case of coupled clinical symptoms, the neurologist will contact the participant for progression monitoring.

2.4.4. Biological samples

The AGUEDA trial will test for several biomarkers using blood, saliva, and fecal samples. Blood samples will be collected by a nurse with participants in fasting conditions (08:00–10:00 a.m.) at the Virgen de las Nieves University Hospital. Briefly, 6 ethylenediaminetetraacetic acid (EDTA) tubes, 4 citrate tubes, 4 serum tubes and one PAXGENE tube will be collected corresponding to a total of approximately 50 ml. Part of the blood samples obtained from each participant will be directly processed at the Hospital (1 EDTA, 1 citrate and 2 serum tubes), and the remaining samples will be stored at −80°C for future studies. Saliva samples will be collected in a falcon tube and a deep oral examination in a subsample will be conducted by an odontologist at the Hospital, recording number of teeth present, clinical attachment loss, and probing depth at a minimum of three sites in at least two teeth per sextant or in all teeth to asses periodontal disease. Fecal samples will be collected in standardized conditions using plastic sterile containers for each participant and then stored at −80°C for future analysis. **Table 5** shows sample type, initial targeted analysis and preliminary analysis plan. Despite this, the analyses will be performed with the best method available at the moment of sample processing. Briefly, blood analytes include

TABLE 3 Cognitive tests included in the AGUEDA trial.

Cognitive test	Session	Format	Domain(s)	Time	Screening	Baseline	Mid-point	Post-intervention	EF score
Spanish telephone interview for cognitive status (Munoz-Garcia et al., 2019)	Telephone	Verbal	GC	10 min	x				
SCD (Jorm et al., 1997)	Telephone	Verbal	SCD	2 min	x			x	
Montreal cognitive assessment (Ojeda et al., 2016)	On-site screening	Verbal-paper-pencil	GC	10 min	x			x	
Mini-Mental State Examination (Blesa et al., 2001)	On-site screening	Verbal-paper-pencil	GC	10 min	x			x	
Rey auditory verbal learning test (Miranda and Valencia, 1997)	1	Verbal	Verbal memory	40 min		x		x	
Trail making test A and B (Reitan, 1958)	1	Paper-pencil	Cognitive flexibility	5 min		x	x	x	x
Digit symbol substitution test (Wechsler, 2008)	1	Paper-pencil	Processing Speed	5 min		x	x	x	x
Rey–Osterrieth complex figure (Rey, 2009)	1	Paper-pencil	Visuospatial memory	20 min		x		x	
Rapid cognitive screening (Malmstrom and Morley, 2013)	1	Verbal-paper-pencil	Cognitive dysfunction	4 min		x	x	x	
Boston Naming Test (Kaplan et al., 2005)	1	Verbal	Language	7 min		x		x	
National Institutes of Health (NIH) toolbox									
Dimensional card sort task (Zelazo et al., 2013)	2	Computerized	EF	5 min		x	x	x	x
List sort working memory (Zelazo et al., 2013)	2	Computerized	Working memory	5 min		x	x	x	x
Picture sequence (Zelazo et al., 2013)	2	Computerized	Working memory	5 min		x	x	x	x
Flanker (Zelazo et al., 2013)	2	Computerized	EF	5 min		x	x	x	x
Wechsler adult intelligence scale IV (Wechsler, 2008)									
Similarities (Wechsler, 2008)	2	Verbal	CI	5 min		x		x	
Vocabulary (Wechsler, 2008)	2	Verbal	CI	10 min		x		x	
Information (Wechsler, 2008)	2	Verbal	CI	5 min		x		x	
Block design (Wechsler, 2008)	2	Verbal	FI	10 min		x		x	
Matrix reasoning (Wechsler, 2008)	2	Verbal	FI	5 min		x		x	
Visual puzzles (Wechsler, 2008)	2	Verbal	FI	5 min		x		x	
Verbal and semantic fluency test (Rosselli et al., 2000)	2	Verbal	Language	4 min		x	x	x	
Spatial working memory (Erickson et al., 2011)	3	Computerized	Working memory	10 min		x	x	x	x
Task-switching	3	Computerized	EF	10 min		x	x	x	x
Stroop task	3	Computerized	EF	10 min		x	x	x	x

SCD, subjective cognitive decline; EF, executive function; CI, crystallized intelligence; FI, fluid intelligence; GC, general cognition.

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TABLE 4 Brain Magnetic Resonance Imaging (MRI) sequences and PET/CT acquisition parameters of the AGUEDA trial.

Measurement	Sequence	Parameters	Acquisition time
MRI	T1-weighted MPRAGE structural	Sagittal, 0.8 mm isotropic resolution, TE/TI/TR = 2.31/1060/2400 ms, FOV = 256 mm, 224 slices	6 min 38 s
	High resolution Hippocampus	Resolution: 0.4 × 0.4 × 2 mm, TE/TR = 79/8830 ms, aligned perpendicular to hippocampus	4 min 53 s
	Resting state EPI	Resolution: 2.5 × 2.5 × 2.5 mm, TE/TR = 40/1000 ms, multiband factor = 8 (CMRR EPI sequence), 64 slices, 480 measurements	8 min 12 s
	fMRI n-back task	Resolution: 2.5 × 2.5 × 2.5 mm, TE/TR = 40/2000 ms, multiband factor = 4, 64 slices, 183 measurements	7 min 12 s
	Diffusion weighted acquisition	Resolution: 2 × 2 × 2 mm, TE/TR = 95.6/2800 ms, multiband factor = 4, *b*-values of 1500, 3000 s/mm2, 64 gradient directions	6 min 18 s
	3D T2 TSE FLAIR	Sagittal, 1 mm isotropic resolution, Turbo spin echo, TE/TI/TR = 388, 2200, 6000 ms, non-selective inversion recovery	7 min 12 s
	pCASL TGSE	3D GRASE pCASL sequence, resolution: 3.1 × 3.1 × 2.5 mm, TE/TR = 23.64/4300 ms, 48 slices, Post-label delay 2 s, background suppression, 10 measurements for labeling and control, four segment readout	5 min 48 s
PET/CT- [90 to 110 min after intravenous injection of [18F]Florbetaben -Neuraceq; Piramal Pharma-]	Computed tomography	400 mA, voltage = 120 kV, 16 s duration, delay time = 2 s, slice thickness = 3 mm, 75 slices, correction = Care Dose, reconstruction = J30s homogeneous/iMAR, FOV = 500 mm	5 min
	Positron emission tomography	Matrix size: 440, activity = 300 MBq (8,1 × 10^{-3} Ci), reconstruction = 10 iterations and 5 subfields, zoom = 2, correction: Attenuation/dispersion/TOF.	20 min

CMRR, center for magnetic resonance research; EPI, echo-planar imaging; FLAIR, fluid attenuated inversion recovery; FOV, field of view; MPRAGE, magnetization prepared rapid gradient echo; pCASL, pseudo-continuous arterial spin labeling; TE, echo time; TI, inversion time; TR, repetition time; TSE, turbo spin-echo; TGSE, turbo gradient spin-echo; VIBE, volumetric interpolated breath-hold examination; PET/CT, positron emission tomography combined computed tomography; mA, milliamperes; kV, kilovoltage; iMAR, iterative metal artifact reduction; Bq, becquerel; Ci, curie; TOF, time of flight.

traditional cardiovascular risk factors, biochemical measurement, inflammatory, brain-peripheral biomarkers, telomere length, and genotyping. In addition, blood and saliva samples will be used to measure amyloid and tau levels. Finally, fecal samples will be used to perform metagenomic analysis of the gut microbiome for each participant.

2.4.5. Physical function

All physical function, muscular strength, and anthropometry will take place at the iMUDS by trained staff (see **Table 6**). Physical function will be assessed using two validated test batteries specifically designed for older adults: The Senior Fitness Test (SFT) (Langhammer and Stanghelle, 2015) and the Short Physical Performance Battery (SPPB) (Guralnik et al., 1994). Briefly, these batteries assess upper and lower body strength, aerobic capacity, walking speed, balance, and flexibility, and while the SFT provides individual information of each component, the SPPB provides a composite score indicating mobility function (Addison et al., 2018). In addition, aerobic capacity will be assessed by the 2-km walking test (Skantz et al., 2020), and gait parameters (e.g., flight times, contact times, and gait variability) by the optical *OptoGait* system (Gomez Bernal et al., 2016).

2.4.6. Muscular strength

Muscular strength will be measured with the handgrip strength test and the upper and lower body isokinetic machine (**Table 6**). In the handgrip strength test, each participant will be encouraged to perform the maximal isometric force twice with each hand using a hand dynamometer (TKK 5101 Grip D, Takey, Tokyo Japan), and the maximum value of each hand will be taken and averaged. Upper and lower body isokinetic strength will be measured using a Gymmex Iso-2 dynamometer (EASYTRCH s.r.l., Italy) according to the previous published protocol (Artero et al., 2012).

2.4.7. Physical activity monitoring

Objectively measured physical activity will be completed using the Actigraph GT3X + accelerometry device (Pensacola, FL, USA), a tri-axial accelerometer, with raw data filtering technology and with a dynamic range of ±6 g (**Table 6**). Each participant will receive an accelerometer and will be required to wear it for 10 days on the non-dominant wrist including periods of sleep, removing it only while bathing or swimming. Along with the device, a diary will be provided in which the participant will record wearing/non-wearing time along with sleep information. Actilife v.6.13.3 will be used to initialize the device with a sampling rate of 100 Hz. To ensure a minimal loss of data, we will perform an automatic check for quality control of the data with the criteria of 3 valid days on week and 1 valid day of weekend on a rolling basis using the GGIR package (Migueles et al., 2019). In the case of non-valid registration, the participant will complete another 10 days of assessment. This measurement will be performed at baseline and again in week 12 of the intervention.

2.4.8. Anthropometric, body composition and blood pressure measurements

For anthropometric and body composition measurements, participants will be asked to arrive at the IMUDs between 08:30 and 10:30 a.m. after overnight fasting. Body weight (kg) will be measured in triplicate with an electronic scale (SECA 861, Hamburg, Germany) with an accuracy of 100 g. Height (cm) will

TABLE 5 Biological samples and targeted analysis plan.

Sample	Analysis target	Preliminary analysis plan
Blood samples		
Plasma	Aβ (Amyloid beta) peptides Aβ1-42, Aβ1-40; T-tau and P-tau; Neurofilament light chain (NFL); Vascular endothelial growth factor (VEGF); chemokine C-X-C motif ligand 13 (CXCL-13); IgM-1; IL-17; pancreatic polypeptide (PPY); Adiponectin; Brain-derived neurotrophic factor (BDNF); Cathepsin B; vascular cell adhesion molecule 1 (VACM-1); C-reactive protein (CRP); Interleukin-6 (IL-6); Interleukin-17 (IL-17); Tumor necrosis factor (TNF-Alpha); Klotho; GLP-1, Clusterin	Plasma biomarker Aβ (Amyloid beta) peptides Aβ1-42, Aβ1-40; T-tau and P-tau; Neurofilament light chain -NFL- concentrations will be measured using Single molecule array (Simoa) methods on an HD-X instrument and commercial assays from Quanterix. Key inflammatory biomarkers analyzed in plasma will analyzed with automated blood cell counters and quantified by multiple analyte profiling technology (MILLIPLEX® MAP Human High Sensitivity T-Cell Magnetic Bead Panel, EMD Millipore Corporation, Missouri, U.S.A.) Klotho will be determined using a solid-phase sandwich enzyme-linked immunosorbent assay (Demeditec, Kiel, Germany) according to the manufacturer's protocol.
Peripheral blood mononuclear cells	Telomere length	A commercial DNA isolation kit (Qiagen, Barcelona, Spain) will be used to extract genomic DNA from isolated PBMCs. Relative telomere length will be determined by quantitative real-time polymerase chain reaction (qRT-PCR) using the telomere/single-copy gene ratio (T/S)
Serum	Insulin-like growth factor 1 (IGF-1); Insulin; Glucose; Glycated Hemoglobin (HbA1c) and Serum fatty acid (SFA-1)	IGF-1, insulin, glucose, glycated hemoglobin (HbA1c) and serum fatty acids (SFA-1) will be analyzed using XMap technology (Luminex Corporation, Austin, TX) and human monoclonal antibodies (Milliplex Map Kit; Millipore, Billerica, MA). Total cholesterol, HDL-C, LDL-C, triglycerides and glucose will be assessed using a spectrophotometer. Insulin will be assessed by chemiluminescence immunoassay.
Whole Blood	miRNA; DNA (genotyped)	Blood specimens will be used for genotyping. The genotypes will be determined using TaqMan genotyping assays. A commercial RNA isolation kit (Qiagen, Barcelona, Spain) will be used to extract total RNA from whole blood samples. The selection of specific miRNAs will be based on the existing bibliographic information. In accordance with this, the selected miRNA will be evaluated using Taqman probes.
Saliva samples		
Saliva	Aβ peptides Aβ1-42, Aβ1-40; T-tau and P-tau	To be determinate with the best method available at the moment of analysis
Fecal samples		
Fecal	Metagenomic Analysis of the Gut Microbiome	To be determinate with the best method available at the moment of analysis

be assessed with a precision stadiometer (SECA 225, Hamburg, Germany) to the nearest 0.1 cm. Head, neck, waist and hip circumference will be measured with an inextensible tape in triplicate and average values will be calculated. Body composition will be measured using dual energy X-ray absorptiometry (DXA, Discovery Densitometer from Hologic). A total body scan as well as specific assessment of both hips and lumbar spine will be performed to determine body composition and bone density parameters. Blood pressure will be measured with patients seated with the left arm resting on a table at heart level and a proper cuff size is fitted. Validated automated blood pressure monitor (Omron M3, Intellisense) will be used to take blood pressure measurements. Three readings will be collected: immediately, following 5 min of rest, and following a subsequent 2 min interval (**Table 6**). Blood pressure is will be calculated as the average of the last two BP readings (Williams et al., 2018).

2.4.9. Psychosocial, mental health and other questionnaires

A battery of questionnaires will be administered including the following dimensions: mental health, psychological health, functional ability, medical information, lifestyle behaviors, and others (**Table 7**). Exercise self-efficacy, social provisions and groups, mood and anxiety, personality traits, social networks, pain, fatigue, etc. have all been shown to either influence exercise adherence or cognition and might explain, mediate, or moderate some of the effects of exercise on cognition (Stillman et al., 2020). As such, we carefully selected a battery of tests that would considerably extend the knowledge on how exercise influences cognitive and brain health.

2.5. Randomization

Randomization will occur on a rolling basis and only after the completion of all baseline assessment sessions by the participant to reduce the risk of bias during the assessment. We will use a computerized randomization protocol through the REDCap Software (Harris et al., 2019) that will incorporate a checklist to make sure all assessments have been completed and the data have been adequately entered into the database. We will use a minimization algorithm with equal allocation (1:1) to one of the two study groups: (1) a 24-week resistance exercise program (3 sessions/week, 60 min/session) group, or (2) wait-list control group which will be asked to maintain their usual lifestyle. The wait-list strategy implies that the control group will also receive

TABLE 6 Physical health parameters.

Parameter	Instrument	Description
Physical function	Senior fitness test (SFT)	Consists of six functional measures of strength in arms and legs, endurance, balance, agility and flexibility. The tests included are, chair stand test, arm curl test, 6-min walk test, 2-min step test, chair-sit-and reach test, and back scratch test.
	Short physical performance battery (SPPB)	Consist of three test to measure functional status. The tests include are, hierarchical assessment of standing balance, usual walking speed over 4 m and five chair sit-to–stand test.
Muscle strength	Hand dynamometer TKK 5101	Each participant is encouraged to perform the maximal isometric force twice with each hand.
	Gymmex Iso-2 dynamometer	Unilateral left and right-side test of upper and lower muscle strength are performed. The subject is instructed to sub maximally flex and extend the knee and elbow 3 times, and then to complete five maximal repetitions.
Gait parameters	OptoGait system	The participant's is encouraged to walk during 5 min to the fast comfortable walking pace in 9 m round trip, safely and without running. The middle 5 m are recorder for analysis. The goal is to record at least 150 steps for each participant.
Body composition	Anthropometric measurement	Body weight and height. Head, neck, waist, and hip circumferences.
	DXA scan	Total body, left and right hips and lumbar spine.
	Neck Magnetic Resonance Imaging (MRI)	An MRI sequence is performed to determinate the neck fat. The sequence parameters are: Resolution: 2 × 2 × 2 mm, TE1-2/TR = 1.23–2.46/5.21 ms, flip angle 9°, CAIPIRINHA iPAT factor = 4, FOV = 448 mm; Duration: 21 s. The patients are instructed to hold breath during the acquisition.
Blood pressure	Omron M3, Intellisense	Three reading will be collected to determine systolic and diastolic blood pressure.
Physical activity	Actigraph GT3X + accelerometry device	The participant's is encouraged to wear the device in the non-dominant wrist during 10 days.

TE, echo time; TR, repetition time; FOV, field of view.

the exercise program after all the post-intervention assessments of the AGUEDA trial have been completed. To improve the representativeness of the study sample and the generalizability of future findings, stratification by two factors will be considered: age at study entry ($\leq$72, >72 years) and sex (male, female). This procedure will be carried out by a blinded researcher completely external to the project (VCS). After the randomization procedure, the sport scientist in charge of the exercise program will be responsible for communicating with the participant about the group allocation.

2.6. Blinding

Due to the nature of the intervention (physical exercise), it is not possible to "blind" the participants, as they will inevitably know their group allocation. However, the principal investigator (PI) and research staff (who do not conduct the exercise sessions) will be blinded to group assignment. Only in the case of an unforeseen event occurring with a participant, the PI will be able to break this blinding condition.

2.7. Resistance exercise program

Detailed description of the exercise program is fundamental to ensure that the program can be disseminated and carried out as clearly and similarly as possible by others, and therefore be replicated in any similar context. Therefore, a detailed and broader description of the program will be further extended in the Consensus on Exercise Reporting Template (CERT) study protocol of the AGUEDA trial (in preparation). Briefly, participants assigned to the 24-week resistance exercise program will be required to attend 3 sessions per week, for approximately 60 min each day, at the Sport Center located at the iMUDS. The intervention will consist of a combination of upper and lower body exercises using elastic bands with different resistance levels and the participants' body weight as the primary resistance. Considering that the participants will be inactive at baseline, we will dedicate the first two weeks of the program to perform exercises for neuromuscular adaptations of both upper and lower body, and for familiarization with the movement patterns and equipment. The exercises will be based on basic movement patterns involving large muscle groups: horizontal traction, vertical traction, horizontal thrust, vertical thrust, hip extension and flexion, hip dominants, knee dominants, anti-rotation, anti-extension, anti-flexion, anti-lateral flexion (Ribeiro et al., 2020). The exercise program is organized into three levels of difficulty, allowing participants to progressively increase the effort intensity while learning and improving their physical performance. In addition, three model sessions have been designed for each of the program levels, and each 60-min session (regardless of the model) will follow the same organization: 1st Warm-up phase (10 min), 2nd Main phase (45 min), and 3rd Cool-down phase (5 min). It is important to

TABLE 7 Psychosocial, mental health and other questionnaires.

Questionnaire	Description
Mental health	
Geriatric Depression Scale (GDS) (Izal and Montorio, 1993)	Measures the presence of depressive symptoms in older adults (30 items).
Hospital Anxiety and Depression Scale (Zigmond and Snaith, 1983)	Questions about anxiety, feelings of depression and emotional distress in the last week (14 items).
Perceived Stress Scale (PSS) (Remor, 2006)	Questionnaire that measures the degree of stress perceived in relation to certain life events during the last month (14 items).
Rosenberg Self-Esteem Scale (RSE) (Robins et al., 2001)	By assessing positive and negative feelings about the self, this scale measures global self-esteem (10 items).
Satisfaction With Life Scale (SWLS) (Vazquez et al., 2013)	It measures the degree of satisfaction with the conditions of their life on a scale of 1 to 7. It measures overall life satisfaction (5 items).
UCLA Loneliness Scale (UCLA-L) (Russell et al., 1980)	This scale assesses feelings of loneliness and social isolation (10 items).
Psychological variables	
Social media use	Measures the frequency and usefulness given to social networks (5 items).
Social Support Questionnaire	It measures the type and quality of social relationships maintained by the subject and the degree of social support received (11 items).
Big Five Inventory (BFI) (Caprara et al., 1993)	It measures the Big five dimensions of personality: Extraversion, agreeableness, conscientiousness, emotional stability, and openness to experience (44 items).
Short Form 36 Health questionnaire (SF-36) (Alonso, 2003)	Questionnaire that measures quality of life according to eight health-related dimensions: physical functioning, physical limitations, pain, behavioral disturbances due to emotional problems, mental capacity, health perception, social functioning and energy/fatigue perception (36 items).
Functional ability variables	
Instrumental activities of daily living (Lawton and Brody, 1969)	Questionnaire assessing daily cognitive and self-care activities in terms of routine and ability (26 items).
International Fitness Scale (IFIS) (Merellano-Navarro et al., 2017)	It subjectively assesses fitness levels by stratifying a population into fitness level groups (5 items).
Mobility and agility questionnaire	Questions about current health status and existing limitations for certain activities (8 items).
Questions about energy and frailty	It measures the energy levels perceived by the subject during the last time, frailty and body weight variation throughout life (9 items)
McGill Pain Questionnaire (Melzack, 1987)	Questionnaire assessing qualitative experiences of pain through affective, sensory, and perceived pain intensity descriptors (16 items).
Medical history	
Health history	Details health history, illnesses, injuries, operations and any other related events that have occurred at any time in life.
Medication list	Information on current and permanent prescription and over-the-counter medications, as well as information on visual impairments.
Oral health questionnaire (Schroder et al., 2011)	Questions about the subject's oral health (10 items)
COVID questionnaire	Questions about COVID-19 information and COVID vaccine status. (40 items)
Adverse event report	Questions about adverse event during the period in the trial participation. (40 items)
Lifestyle behaviors	
Drinks questionnaire	Questions about the frequency of intake of coffee and alcoholic beverages (14 items).
Food-frequency Questionnaire (FFQ) (Boylan et al., 2009)	Subjects are asked to indicate the frequency and usual amount of food and beverage consumption using response categories (148 items).
MEDAS-14 (Schroder et al., 2011)	Questionnaire of adherence to the Mediterranean diet (14 items).
Smoking questions (Schroder et al., 2011)	Questions on smoking habits throughout life (4 items).
Sitting time questionnaire	Subjects respond how much time they spend sitting at work, at home, while watching TV or in their free time, and driving on a typical day during weekday and weekends. (8 items)
Physical activity assessment	Questions on the perception and performance of physical activity during life (77 items).
Pittsburgh Sleep Quality Index (PSQI) (Buysse et al., 1989)	Survey that measures the sleep quality (21 items).
Others	
Demographic questions (Schroder et al., 2011)	Questions about the family's economic situation, years of education, and doctor visits (16 items).
Edinburgh Handedness Inventory (EHI) (Oldfield, 1971)	Questions about how often the subject uses one or the other hand for different tasks (9 items).

note that the maximum training ratio is 1/6, i.e., there can be a maximum of 6 participants per trainer in each session. Load and intensity of the exercises will be defined by: (i) number of repetitions for each set; (ii) elastic band resistance in ascending order with seven intensities (i.e., yellow to gold (Uchida et al., 2016); (iii) difficulty of exercises (i.e., 3 levels); and (iv) time of performance following the "as fast as possible" indication for every repetition.

The prescribed intensity will be reached at 70–80% of the Rating of Perceived Exertion (RPE) (Colado et al., 2014). The RPE is a scale from 1 to 10 (i.e., from very mild to extremely maximal effort) by which participants will be able to indicate their own perception of the effort made during each exercise. This scale will be visible to participants throughout the session. In addition, the intensity will increase progressively, and the adjustment of each participant's working loads will be performed using a target RPE, modifying execution time and cadence, and the elastic band type. Heart rate will be monitored during all sessions through a Polar H10 monitor with mobile phone applications. Data will be logged second-by-second into the Polar Beat app and Elite HRV which will be linked to the sensor band. In addition, sleep quality and pre- and post- affective response will be recorded in each session (Hardy and Rejeski, 1989). The exercise program will be conducted by a sport science professional with a bachelor degree in Physical Activity and Sports Sciences or master degree in Personal Training. Adherence to the resistance exercise program protocol will be evaluated as session attendance at the program, and a minimum of 80% attendance to all exercise sessions will be required to meet the protocol. To facilitate adherence, when any participant misses an exercise session for any reason, they will have the possibility to attend another exercise group to make up the missed session. Reasons for non-attendance and details about each recovered session will be documented. The sport science professional in charge of the program will encourage the participant to attend all exercise sessions and will be responsible for rescheduling in case of necessity. The control group will be asked to maintain its usual lifestyle and will receive the same resistance exercise program after the post-intervention assessments at week 24. In order to favor the adherence and the interaction with the control group, bimonthly activities will be performed consisting of meetings in which participants play games, engage in relaxing activities, among others. Adherence strategies will be implemented during the intervention program to minimize potential losses, including birthday messages, games-based and social meetings, among others. In both groups, participants will continue with concomitant care and activities. Participants will be invited to the testing at post-intervention irrespectively of their compliance to the exercise intervention protocol.

2.8. Adverse events and participant safety

Because the effects of resistance exercise are well tolerated, minor unrelated non-serious adverse events (e.g., allergies, upset stomach) and minor adverse events related to the practice of physical exercise (e.g., myalgia, sweating, and dyspnea) will not be recorded, unless it requires a change of the intervention (e.g., change in a specific exercise due to articular pain). Any situation that may occur, such as injury, emergency or scheduled surgery, will be reported and evaluated by the research team and the PI will be unblinded when necessary. The adverse event will be recorded in a custom form with detailed information of its seriousness, severity, chronicity, and resolution. Thus, adverse events as well as any possible effects produced by the assessments or the exercise intervention will be carefully monitored, recorded and reported in REDCap. The severity of the adverse event will be determined using common terminology criteria for adverse events (i.e., mild, moderate, severe) (Freites-Martinez et al., 2021), and any adverse event rated category three will be considered as a serious adverse event. If the situation is considered unsafe to continue in the study, temporary or definitive exclusion from the project will be discussed and reported. In the case of temporary exclusion (e.g., any joint lesion) the last exercise session will be recorded, and a physician clearance will be required to rejoin to resistance exercise program.

2.9. Data management and dissemination policy

The AGUEDA trial will use two data storage systems: (i) the REDCap platform is an online platform designed to store and manage electronic data and create research databases for clinical trials and translational research. The AGUEDA trial will use this platform for the storage and management of all non-imaging data (e.g., behavioral or cognitive data). This platform will also contain the data entry forms that can be downloaded and printed if necessary. The database of the project will be generated directly from REDCap; (ii) the AGUEDA desktop computer will also contain all participants' data collected on paper-based forms, that will be scanned and stored, and the MRI and PET image data. It is important to note that storing all the data in REDCap ensures that all the information is safe in case the server is damaged for any reason (Harris et al., 2019). In addition, the database will be automatically stored once a week on a remote hard drive with the data collected and registered at the moment of storing. All physical copies of data collection forms and documents will be stored in a locked cabinet also located at the iMUDS. From the moment participants are enrolled in the trial, we will assign them a code that will be used in all forms and data collection reports, ensuring anonymity of all data. The PI will have continuous access to the final dataset. Participants authorize in the informed consent that their pseudonymized data may be made available for secondary research importantly, once the participant is enrolled and before randomization, all data will be counted, checked for quality control purposes, and archived following the storage protocol described above by the research team, performing these same steps at the three measurement points. Specifically, quality control includes visual immediate inspection of imaging data (MRI and PET/CT), double correction of paper-pencil cognitive test and check limit of real data (ranges and valid type and values) in REDCap.

Data access will operate under the findability, accessibility, interoperability, and reusability (FAIR) principles, and code of analysis will be made publicly available. We will consider governance, ethical and shared trial oversight, and expectations

that trials will adhere to the best practice of the day. The results of this study, whether positive, negative, or inconclusive, will be published in peer-reviewed journals as well as at national and international conferences. The study findings will be also released to the participants, physicians, and the general medical community through social media and press. Authorship eligibility will be handled by the PI according to the International Committee of Medical Journal Editors and all authors will contribute to the final revision of each manuscript.

2.10. Power and analytic strategy

To calculate the sample size of the AGUEDA trial, we based our estimates on the AGUEDA primary outcome, i.e., change on the executive function composite score over a 24-weeks resistance exercise period. To calculate the sample size, we based the power analysis on a meta-analysis which showed that the effect size of exercise intervention on executive function in older adults was 0.34 (95% Confidence interval = 0.22 to 0.47) (Northey et al., 2018), with a two-tailed alpha at 0.05 and a power of 80%. Thus, we need a sample size of 35 participants for each group to obtain such an effect size. If the maximum dropout rate is 20% plus some residual power, we then need 45 participants for each group. After all computations, we decided that the target sample size for this study will be 90 participants.

To investigate the overall effect of the resistance exercise intervention on the change from baseline to post-intervention in the executive function composite score, a general linear mixed model (GLMM) approach with repeated measures over time (baseline and post-intervention outcomes) will be used. This statistical mixed model will include both a random intercept and slope individually for each participant and a group-by-time interaction as a fixed effect. Analyses will be primarily performed according to the intention-to-treat principle (including all participants randomized) and, in addition, based on a per-protocol principle (i.e., including only those participants with valid data for baseline and post-intervention assessments and an attendance rate to exercise program $\geq$ 80% of all sessions). To test the effect of the AGUEDA trial on the secondary cognitive outcomes (i.e., general cognition, verbal memory, visuospatial memory, language, and intelligence), the same analyses using an intention-to-treat principle, and GLMM analysis will be performed. All other analyses related to brain structure and function, brain amyloid load, blood biomarkers, genetics, body composition, physical function, mental health, questionnaires, among others are considered secondary or tertiary outcomes and for the mediation and moderation analysis. The plan is to test these outcomes in a similar manner as described above using an intention-to-treat principle and a GLMM approach using a group x time interaction. In addition, the mediation and moderation analysis will be performed following AGReMA (A Guideline for Reporting Mediation Analyses) recommendations (Lee et al., 2021), using the group as independent variable (wait-list vs. intervention group), executive function as outcome, and mediating and moderating variables included will be based on three levels of mechanisms proposed in the literature (i.e., molecular, brain structure and function, and behavioral levels) (Stillman et al., 2020). The potential effects of missing data will also be explored through various imputation models and sensitivity analyses. Thus, to fully appreciate the potential influence of missing data, we will perform sensitivity analyses to examine whether the imputation method changes the results. No interim analysis will be performed.

3. Discussion

Although exercise has the potential to improve brain health (e.g., brain structure and function) in older adults (Stillman et al., 2020; Liang et al., 2021), there is still scarce knowledge on what dose of exercise (i.e., type, duration, frequency and intensity) is sufficient for improving brain health. Well-designed intervention studies on exercise effects on brain health may also help to determine the feasibility of different types of exercise for brain guidelines, including resistance exercise. Further research in this area will help to provide rigorous, evidenced-based recommendations on how resistance exercise can facilitate improvements in brain and cognitive health.

3.1. Expected results

To our knowledge, only few studies has evaluated the efficacy of a resistance exercise program on brain health in cognitive healthy older adults, with a lack of including a broad understanding of several cognitive (e.g., executive function) and biological (e.g., brain Aβ) Alzheimer-related outcomes as well as its mediators/moderators. The main hypothesis is that our resistance exercise program will improve executive function, showing at least a 0.34 effect size in comparison to the wait-list control group. In addition, we also predict that molecular, brain structure and function, and behavioral changes will mediate the effect of the exercise program on executive function. Further, while there is growing evidence on the individual effects of resistance exercise on cognitive and brain health indicators, some of the expected contributions of this work to the field is the fundamental need to (i) integrate both its direct effects on cognitive indicators together with its indirect effects at the three proposed levels of mechanisms, (ii) test the interdependence of molecular-brain-behavioral mechanistic levels and, (iii) elucidate its bidirectional effects, such as feedback loops, that are likely to exist between levels.

3.2. Gaps and mechanisms related to the main hypothesis

Physical exercise is one of the most relevant non-pharmaceutical approaches to prevent several chronic and progressive diseases, including AD (Gleeson et al., 2011; Fiuza-Luces et al., 2013; Liang et al., 2022). While research about the effect of exercise interventions on brain health is increasing, several questions about type and dose remain inconclusive. Particularly, combined evidence on the effect of resistance exercise interventions on executive function and its mechanisms at

molecular, brain structure and function and behavioral levels is largely unknown. However, isolated evidence might support the premise of our hypothesis. First, resistance exercise has shown to improve cognition, being an alternative to other types of exercises (Chang et al., 2012; Li et al., 2018; Northey et al., 2018; Landrigan et al., 2020; Gallardo-Gomez et al., 2022). Second, at a molecular level, resistance exercise is a key factor in the secretion of BDNF (Marinus et al., 2019) and IGF-1 levels in older adults, (Jiang et al., 2020) and acutely in irisin levels of young adults (Tsuchiya et al., 2015). Despite this, there is still a need for evidence on resistance exercise interventions since this type of exercise might additionally boost certain exerkines (Vints et al., 2022). Interestingly, resistance exercise in animal models triggers specific epigenomic changes of interferon pathways in the hippocampus, related to interleukin-1 beta expression, which plays a role in the inflammatory response of the brain (Urdinguio et al., 2021), and reduces neuroinflammatory cytokine expression (Kelty et al., 2022). Third, at structural and functional brain levels, only a few studies have investigated the effect of resistance exercise programs on brain structure and function in older adults. Specifically, previous evidence shows that resistance exercise influences brain volumes and morphology (Best et al., 2015; Kim et al., 2017), specifically hippocampal volumes (Broadhouse et al., 2020), white matter lesions (Bolandzadeh et al., 2015) and atrophy (Best et al., 2015), cortical thickness and functional connectivity (Suo et al., 2016), while other measures such as cerebral blood flow are understudied (Bliss et al., 2021; Smith et al., 2021). Despite this, some observational studies show that resistance exercise participation is related to cerebral blood flow in older adults (Xu et al., 2014), and there is evidence of changes in functional connectivity and cerebral blood flow in other populations (Akbar et al., 2020; Thomas et al., 2021; Huang et al., 2022). On the other hand, no previous human studies have examined the effects of resistance exercise on specific (i.e., brain Aβ) Alzheimer-related biomarkers, and few studies examined the effect on peripheral (i.e., Aβ or Tau) biomarkers (Baek et al., 2021; Hyang-Beum and Tae-Sang, 2021), with no conclusive results. Finally, at the behavioral level, resistance exercise shows effects on a broad array of mental health outcomes in older population, such as health-related quality of life (Hart and Buck, 2019), depressive (Moraes et al., 2020; Cunha et al., 2021; De Sousa et al., 2021) or anxiety symptoms (Cunha et al., 2021), and sleep (Kovacevic et al., 2018; Li et al., 2021), but the study of these measures as potential mediators of cognition are needed.

3.3. Limitations

Our study will provide considerable advances in the knowledge of resistance exercise effects on brain and cognitive health. However, there are some limitations that we must acknowledge. We did not perform an extensive cognitive adjudication to clinically identify cognitive status. Instead, we used STICS-m, MoCA and MMSE tests to determine cognitive status. While these are classical cognitive tests commonly used in clinical and research settings, they may have limitations in terms of sensitivity and specificity for detecting MCI. To address this limitation, we applied important education/age adjustment to secure the highest specificity in this specific population. In addition, we selected a wait-list control group, we will not be able to directly demonstrate that a resistance exercise program is more effective than other types of exercise in this population. The main goal of this RCT is to demonstrate that resistance exercise is also a type of exercise to improve cognition in older adults and to understand its underlying mechanisms. In addition, findings from the AGUEDA trial should be interpreted in the context of the sample characteristics, namely, cognitively normal older adults without physical limitation to perform exercise, and it is not possible to extrapolate to other populations.

An overarching limitation in the field is that the details of how exercise interventions are implemented are often poorly reported. Detailed prescriptions and exercise intervention characteristics are necessary to translate into clinical practice (Gallardo-Gomez et al., 2022; Hansford et al., 2022). The AGUEDA trial will overcome this limitation and will deliver a detailed and broader description of the program to generalize the prescription of resistance exercise in an older population. Importantly, one of the strengths of the AGUEDA exercise program is the minimal equipment requirement, which enables the application and translation of the results to different contexts and clinical practices. Lastly, another important characteristic of our study is the assessment of a broad spectrum of physical (e.g., fitness, body composition, and gait variability), brain (e.g., executive function, brain structure and function and mental health) and behavioral (e.g., sleep, diet, and sedentary time) indicators to fully understand mediators and moderators of resistance exercise-induced cognitive changes in older adults.

4. Conclusion

The AGUEDA trial will shed light on the effect of a 24-week resistance exercise intervention on executive function, and to understand its underlying mechanisms. The results of the AGUEDA trial will have important clinical implications given the increase of older adults in the population and the prevalence of cognitive decline in this age range. The AGUEDA trial will also help to establish an effective preventive strategy based on resistance exercise for delaying cognitive decline and dementia.

Ethics statement

The trial protocol is in accordance with the principles of the Declaration of Helsinki and has been approved by the Research Ethics Board of the Andalusian Health Service (CEIM/CEI Provincial de Granada; #2317-N-19).

Author contributions

IE-C conceived the study. IE-C, PS-U, and CM-H designed the study. PS-U, CM-H, MG-R, KE, AC, FO, and JM-G

made substantial intellectual contribution to design of the study. PS-U, CM-H, CC-R, BF-G, MO-R, AC-P, and AT made substantial contributions preparation of material and protocols. PS-U, CM-H, YG-R, and ET-I substantial logistical contributions for the acquisition of data. All authors contributed intellectually and approved the final manuscript of this study protocol.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fnhum.2023.1168549/full#supplementary-material

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AGUEDA Project

Active Gains in brain Using Exercise During Aging

Statistical Analysis Plan

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1. General information of AGUEDA project

1.1 Background and rationale

Cognitive decline associated with dementia, particularly in Alzheimer's disease (AD) (1), initiates over two decades before clinical symptoms, with pathological accumulation of beta-amyloid in the brain as an early marker (2,3). Executive function and memory are the first cognitive domains affected (4). While memory decline is typical of AD, executive dysfunction may precede it (5) and predict further cognitive decline, mild cognitive impairment (MCI), and AD, making it a potential target for lifestyle interventions (6,7).

Dementia is not inherent to aging, and exercise stands out as a non-pharmacological preventive and efficient treatment for age-related cognitive decline (8). Systematic reviews have highlighted the benefits of exercise interventions in older adults, impacting executive function, memory, language, and general cognition (9–12). Although aerobic exercise has been extensively studied (13,14), resistance exercise, despite being included in physical activity guidelines, has received less attention (15,16). Recent reviews suggest significant cognitive benefits from resistance exercise (12,17,18), even at low doses (18). However, variations in study parameters make it challenging to determine the specific dose and type of resistance exercise needed for cognitive improvement (19). Therefore, well-designed randomized controlled trials (RCTs) are essential to clarify the effects of resistance exercise on cognition in older adults.

Potential mechanisms underlying exercise's cognitive benefits span molecular, brain structural/functional, and behavioral levels. Most evidence is derived from aerobic exercise studies, but resistance exercise may involve distinct pathways. Animal studies suggest inflammatory and metabolic changes, while human studies highlight neurochemical markers. Understanding these mechanisms, especially in cognitively normal older adults, requires well-designed RCTs with clear reporting of exercise characteristics.

1.2 Objective

The main objectives are:

1. The primary aim of the study is to investigate the effects of a 24-week resistance exercise program on executive function in cognitively normal older adults.
2. The secondary aims are (i) to examine the effects of resistance exercise on brain structural and functional markers (e.g., grey and white matter measures, functional connectivity or cerebral blood flow), brain Aβ deposition, peripheral molecular markers (e.g., BDNF, IGF-1, and Aβ), and other cognitive outcomes (e.g., memory and language) and (ii) to investigate mediators and moderators of the potential exercise-derived improvements observed in executive function and brain markers.
3. The tertiary aim is to investigate the effect effects of a 24-week resistance exercise program on other outcomes including changes in muscular strength, cardiorespiratory fitness, physical function, gait parameters, mental health, quality of life, diet, body composition, blood pressure, microbiome, among others.

2. Study methods

2.1 Trail design

The AGUEDA (Active Gains in brain Using Exercise During Aging) trial is a single-centre, two-arm, single-blind RCT in which 90 cognitively normal older adults, aged 65–80 years old are randomized into a resistance exercise group (n = 45) or a wait-list control group (n = 45). Measurements are performed at baseline, at midpoint (12 weeks) and post intervention (24 weeks). A visual representation ofthe participant flow during the study can be seen in **Figure 1**. More detailed information is described in the study protocol paper (20). The study has been designed following the Standard Protocol Items for Randomized Interventional Trials (SPIRIT)(21).

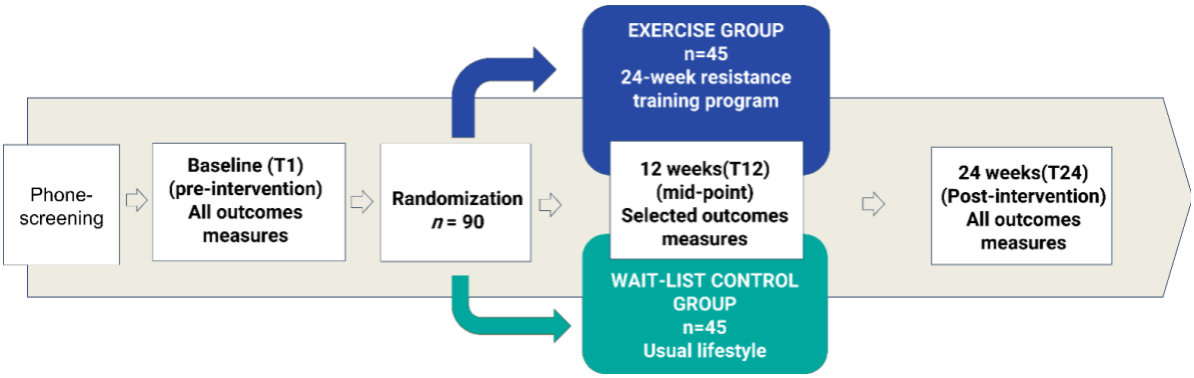


Figure 1. Overview of the AGUEDA project

2.2 Randomization

Randomization (1:1) occurs on a rolling basis and only after the completion of all baseline assessment sessions by the participant to reduce the risk of bias during the assessment. We use a computerized randomization protocol (stratified by age and sex) through the REDCap Software (22) that incorporate a checklist to make sure all assessments have been completed and the data have been adequately entered into the database.

2.3 Sample size

We based our estimates on the AGUEDA primary outcome, i.e., change on the executive function composite score over a 24-weeks resistance exercise period. The power analysis is based on a meta-analysis which showed that the effect size of exercise intervention on executive function in older adults was 0.34 (95% Confidence interval = 0.22 to 0.47) (17) with a two-tailed alpha at 0.05 and a power of 80%. Thus, we need a sample size of 35 participants for each group to obtain such an effect size. If the maximum dropout rate is 20% plus some residual power, we then need 45 participants for each group. After all computations, we decided that the target sample size for this study will be 90 participants. More detailed information is described in the protocol paper (20)

2.4 Framework

A superiority hypothesis testing framework is used, emphasizing the evaluation of a intervention's positive impact compared to a control group. In the main analysis, we compare

resistance exercise group versus control group (no exercise) in 3-time points (Baseline, 12-week and 24-week).

2.5 Statistical interim analyses and stopping guidance

No pre-specified interim analyses are performed, and therefore, a stopping guidance is not applicable.

2.5 Timing of the final analyses

The main final analyses will be performed after the finalization of the data processing of the primary outcome.

2.6 Timing of outcome assessment

The primary outcome is assessed at baseline, midpoint (12 weeks) and post intervention (24 weeks). In addition, there are other secondary outcomes only assessed at baseline and post intervention. The time frame of each outcome is explained in section *5.1 Outcome definitions*.

3. Statistical principles

3.1 Confidence and P values

All statistical tests are two-tailed. *P* for significance will be set at 0.05 and 95% confidence intervals will be estimated.

3.2 Adherence and protocol deviations

Adherence to the exercise program is measured by session attendance which is the proportion of sessions attended out of the total offered sessions.

All protocol deviations made to the protocol (e.g. change in pre-defined inclusion/exclusion criteria, baseline and post assessments, data cleaning/processing) will be reported and described in the analysis.

3.3 Analysis populations

Three analytic datasets will be defined for the statistical analyses:

1. Intention-to-treat: This dataset will be used for our primary analyses and includes all randomly assigned study participants. With this approach, all randomized participants are included in the main analysis.
2. Per protocol: This dataset will be used for our secondary analyses and includes all participants who followed the attendance rate. As such, as the 100% of the session corresponding to a total of 72 sessions (3 session per week for 24 weeks), 80% attendance is required for be included per protocol analyses, corresponding to >= 57 exercise sessions.

4. Trail population

4.1 Recruitment

We began recruitment of community-dwelling older adults in Granada (Spain) city and surrounding areas in March 2021. Strategies to achieve the targeted sample size mainly include mass mailings and social media advertisements, word of mouth, with augmentation of the recruitment strategy by using advertisements in newspapers (TV, radio, and internet).

Information necessary for the CONSORT flow diagram is collected. For the enrollment phase, we note the number of participants that were assessed for eligibility by the research team, the number of excluded participants (plus reason for exclusion), and the number of randomized participants. For the allocation, the number of participants allocated to the intervention and control group is noted. For follow-up, the number of participants lost to follow-up (12-week and 24-week) and the number who discontinued the intervention (plus reasons) are counted. Finally, the number of participants included in the analyses using the intention-to-treat and per-protocol databases is described, along with the reasons for exclusion.

4.2 Initial phone screening

Following recruitment strategies and after the participant's first contact for general information, a phone screening is performed with those individuals who are potentially interestedin participating. Then, participants meeting the phone criteria are invited to the on-site screening, and those who met the on-site criteria are enrolled in the study.

4.3 Eligibility

Inclusion and exclusion criteria is defined as follows: (i) older adults between 65 - 80 years, (ii) physically inactive (i.e., defined as not participating in any resistance exercise program in the last 6 months or accumulating less than 600 METs-Min/week by the International Physical Activity Questionnaire (IPAQ) (23), (iii) classified as cognitively normal according to the Spanish version of the modified Telephone Interview of Cognitive Status (STICS-m) (≥ 26 points) (24), Mini-Mental State Examination (MMSE) (25/30) (25) and Montreal Cognitive Assessment (MoCA) (<71 years, 24/30, 71-75 22/30, >75, 21/30) (26); and (iv) without significant depressive symptoms at baseline according to the Geriatric Depression Scale (GDS) (15) (27). Detailed information about the study design is available elsewhere(20).

4.4 Withdrawal/follow-up

The number, timing, and reasons (e.g. adverse events, withdraw consent, lost to follow-up) for withdrawal is noted and described.

4.5 Baseline patient characteristics

A baseline data table is created to describe the characteristics of the study population. The main characteristics include age, sex, education, body mass index and the outcomes variables. The characteristics of the total study population and each study arm is summarized using mean (SD) or median (interquartile range) for normally and not normally distributed continuous variables, respectively, and as number (percentage) for categorical variables.

5. Analysis

5.1 Outcome definitions

The primary outcome is assessed at baseline (before randomization), at mid-point (12-week) and after the intervention (24-week) while some of secondary outcomes are assessed only at baseline and post intervention (24-week, Table 1).

Table 1. Time frame and specific outcomes of cognitive tests.

Executive function test	Baseline	12-week	24-week	Specific outcome
Trail making test A and B	x	x	x	Interference score
Digit symbol substitution test	x	x	x	Total number of correctly paired symbols
Dimensional card sort task	x		x	Inverse efficiency score
Spatial working memory	x	x	x	Inverse efficiency score
Picture sequence	x		x	Theta score
List sort working memory	x		x	Raw score corrects trials
Flanker	x		x	Incongruent trials inverse efficiency score
Stroop task	x	x	x	Interference score
Task-switching	x	x	x	Global Cost
Other cognitive test	**Baseline**	**12-week**	**24-week**	**Specific outcome**
Rey auditory verbal learning test	x		x	Total recorded words
Rey-Osterrieth complex figure	x		x	Correct drawn elements
Boston Naming Test	x		x	Correct responses
Wechsler adult intelligence scale IV ï Similarities ï Vocabulary ï Information ï Block design ï Matrix reasoning ï Visual puzzles	x		x	Correct responses
Verbal and semantic fluency test	x	x	x	Correct responses

Primary outcome

The primary outcome of the AGUEDA project is the change on the executive function Executive function composite score (Z-score) which will be created by performed confirmatory factor analyses (CFA). CFA will include the executive function variables and the final model will be choose based on the fit of the model considering χ^2 p-value ≥ .05, Bentler Comparative Fit Index (CFI) > .95, Steiger-Lind Root Mean Squared Error of Approximation (RMSEA) < .08 as a good model fit (28).

The executive function tests included to perform the CFA will be:

- Trail Making Test (29). The difference in the time between trail A and trail B (TMT B-A) is calculated and the inverse variable will be used.
- Design symbol substitution test (30). The variable to use is the total numbers of correct symbols score.
- Dimensional Change Card Sort Test (31). The variable used will be the inverse efficiency score of color trials (Reaction time/ Accuracy).
- Spatial working memory (13). Performance is measured based on accuracy, response times, and errors. The variable used will be the inverse efficiency score (Reaction time/ Accuracy) of all trials (2,3,4 items).

- Picture Sequence Memory Test (31). The participant's score is derived from the cumulative number of adjacent pairs of pictures remembered correctly over three learning trials, the theta score.
- List Sort Memory Test (31). Scores are calculated by summing the total number of correct responses across both lists.
- Flanker (31). The interference from reporting the direction of the middle arrow and not the external arrows will be calculated by subtracting the average incongruent accuracy from that of the congruent trials.
- Stroop test. The interference score from the incongruent trials will be calculated.
- Task Switching (32). The difference in mean reaction time for the mixed block of trials, including both the repeat and switch trials, is calculated as a global cost score.

Secondary outcomes

Other cognitive outcomes. Cognitive outcomes are assessed such as general cognition, language, verbal memory, visuospatial memory and crystallized and fluid intelligence.

- Rey auditory verbal learning test (33). The Total Learning measure (RAVLT-AT) will be the capacity to recall and to accumulate words through the 5 learning trials.
- Rey-Osterrieth complex figure (ROCF) (34). The study variables that will be used are: (a) copy—overall sum of correctly drawn elements when copying the model; (b) immediate recall—total sum of correctly drawn items after 3 min.
- Boston Naming Test (BNT) (35). The variable that we will use will be the number of correct responses generated requested to the name of each figure.
- Wechsler adult intelligence scale IV (36). That tests represent a different cognitive domain: Verbal Comprehension, Working Memory Index, Perceptual Reasoning Index and Processing Speed Index. The variable that we will use will be the number of correct responses generated in each test:
  - Similarities
  - Vocabulary
  - Information
  - Block design
  - Matrix reasoning
  - Visual puzzles
- Verbal and semantic fluency test(37). The verbal fluency assessment tests letter fluency (recall as many words beginning with a specific letter) and semantic fluency (recall as many members of a category as possible). The variable that we will use will be the number of words generated by the participant.

Brain structure and function (Time frame: Baseline and 24-week). Brain structure and function is assessed by a Siemens Magnetom PRISMA Fit 3T scanner with a 64-channel head coil. Briefly, the following sequences are performed:

- T1-weighted Magnetization-Prepared Rapid Acquisition with Gradient Echo (T1-weighted MPRAGE structural): Volume, surface, thickness and shape.
- High resolution hippocampus: High resolution Hippocampal subfield segmentation.

- Resting State Echo-Planar Imaging (EPI): resting state Blood Oxygen Level Dependent (BOLD) fluctuations, activation and functional connectivity.
- Functional Magnetic Resonance Imaging n-back Task (fMRI n-back task): Task-evoked BOLD fluctuations, activation and functional connectivity.
- Diffusion weighted acquisition: White matter integrity indicators such as fractional anisotropy, mean diffusivity, radial diffusivity, and axial diffusivity.
- Three-Dimensional T2 Turbo Spin Echo Fluid Attenuated Inversion Recovery (3D T2 TSE FLAIR): White matter lesions.
- Pseudo-Continuous Arterial Spin Labeling Turbo Gradient Spin Echo (pCASL TGSE): Blood flow estimates and arterial transit time estimates.

*Brain amyloid beta deposition (*Aβ) (Time frame: Baseline and 24-week). The brain Aβ deposition is assess using a PET scan Biograph-Vision 600 Edge Positron emission tomography/Computed tomography (PET/CT) digital scanner (Siemens, Erlangen, Germany). [18F]Florbetaben is injected intravenously into each participant in accordance with the applicable regulatory guidelines. PET images are acquired from participants 90 to 110 min after intravenous injection of 300 megabecquerel (MBq) ± 20% (38). The Standardized Uptake Value Ratio (SUVR) and centiloid values will be used as outcomes.

Other outcomes

Biological samples (Time frame: Baseline and 24-week). Several biomarkers using blood, saliva, and fecal samples are determined . Blood samples are used to determine traditional blood markers and genotyping and novel blood brain-based biomarkers. In addition, fecal and saliva samples are used to determine microbiome composition and others potential biomarkers.

Physical Function (Time frame: Baseline, 12-week and 24-week; gait parameters and 2-km walking test not performed at 12-week). The Senior Fitness Test (SFT) (39) and the Short Physical Performance Battery (SPPB) (40) ,the 2-km test (only baseline and 24-week) and the Optogait are performed. Briefly, we measure (i) aerobic capacity [i.e., 6-minute walk test (m/6 min) (41), Two-minute step test (n/2 min), Up and Go test (sec) and 2-km walking test (seconds) (42), Chair Sit and Reach Test (cm)], (ii) balance [i.e., Balance test (score of Balance test)], (iii) flexibility [i.e., Chair Sit and Reach test (cm) and the Back Scratch test (cm)] and (iv) gait parameters [i.e., Stride length (cm), Stride time (sec), Step length (cm), Step time (sec))] by the optical Opto Gaitsystem (Microgate Slr; Bolzano, Italy) (43).

Muscular strength (Time frame: Baseline, 12-week and 24-week; Gymmex Iso-2 dynamometer not performed at 12-week). Measurements are taken to assess both lower and upper muscular strength. Lower body strength was evaluated using the Chair Stand test from SFT (44) and the Gymmex Iso-2 dynamometer (EASYTRCH s.r.l., Italy) (45). Upper body strength was assessed through the Arm Curl test from SFT (44), Hand dynamometer (TKK 5101 Grip D, Takey, Tokyo Japan) and the Gymmex Iso-2 dynamometer (EASYTRCH s.r.l., Italy) according to the previous published protocol (45).

Physical activity monitoring (Time frame: Baseline and 12-week). Movement behaviors are monitored using the Actigraph GT3X + accelerometry device (Pensacola, FL, USA). Participants wear the device on their non-dominant wrist for a continuous span of 9 days, with removal only during activities such as showering. Alongside the device, participants maintained a diary to record in-bed and out-of-bed times, naps, and periods when the device is not in use.

Anthropometric, body composition and blood pressure measurements (Time frame: Baseline and 24-week). Body weight (kg) measured in triplicate with an electronic scale (SECA 861, Hamburg, Germany). Height (cm) assess with a precision stadiometer (SECA 225, Hamburg, Germany). Head, neck, waist and hip circumference are measured. Body composition is measure using dual energy X-ray absorptiometry (DXA, Discovery Densitometer from Hologic). A total body scan as well as specific assessment of both hips and lumbar spine are performed to determine body composition and bone density parameters. Validated automated blood pressure monitor (Omron M3, Intellisense) is used to take blood pressure measurements.

Psychosocial, mental health and other questionnaires. A battery of questionnaires is administered including the following dimensions: mental health, psychological health, functional ability, medical information, lifestyle behaviors, andothers. Exercise self-efficacy, social provisions and groups. mood and anxiety, personality traits, social networks, pain, fatigue, etc. As such, we carefully selected a battery of tests that would considerably extend the knowledge on how exercise influences cognitive and brain health. The time frame of each questionnaire is present in table 2.

Table 2. Time frame of questionnaires

Questionnaire	Baseline	12-week	24-week
Mental health			
Geriatric Depression Scale (GDS)	x		x
Hospital Anxiety and Depression Scale	x	x	x
Perceived Stress Scale (PSS)	x		x
Rosenberg Self-Esteem Scale (RSE)	x		x
Satisfaction With Life Scale (SWLS)	x		x
UCLA Loneliness Scale (UCLA-L)	x		x
Psychological variables			
Social media use	x		x
Social Support Questionnaire	x		x
Big Five Inventory (BFI)	x		x
Short Form 36 Health questionnaire (SF-36)	x	x	x
Functional ability variables			
Instrumental activities ofdaily living	x		x
International Fitness Scale (IFIS)	x		x
Mobility and agility questionnaire	x	x	x
Questions about energy and frailty	x	x	x
McGill Pain Questionnaire	x	x	x
Medical history			
Health history	x		x
Medication list	x		x
Oral health questionnaire	x		x

COVID questionnaire	x		x
Adverse event report	x		x
Lifestyle behaviors			
Drinks questionnaire	x		x
Food-frequency Questionnaire (FFQ)	x		x
MEDAS-14	x	x	x
Smoking questions	x		x
Sitting time questionnaire	x		x
Physical activity assessment	x		x
Pittsburgh Sleep Quality Index (PSQI)	x		x
Others			
Demographic questions	x		x
Edinburgh Handedness Inventory (EHI)	x		x
Mental health	x		x
Geriatric Depression Scale (GDS)	x		x
Hospital Anxiety and Depression Scale	x		x
Perceived Stress Scale (PSS)	x		x
Rosenberg Self-Esteem Scale (RSE)	x		x
Satisfaction With Life Scale (SWLS)	x		x
UCLA Loneliness Scale (UCLA-L)	x		x
Psychological variables	x		x
Social media use	x		x
Social Support Questionnaire	x		x
Big Five Inventory (BFI)	x		x
Short Form 36 Health questionnaire (SF-36)	x		x
Functional ability variables	x		x

5.2 Analysis methods

The main analyses will consist of the intention-to-treat analyses for the primary and secondary outcomes using a constrained linear mixed model (cLMM). The model will include fixed effects for time (three levels or two levels, depending of the outcome) and treatment (two levels) as well as the unique participant identifier as a random effect. Model assumptions will be checked. In case we detect model violations, we will take appropriated measures by, for example, conducting data transformations. The treatment effects will be presented by estimated marginal means and standard errors for changes. The specific differences between analysis are described in the following's points divided into primary analysis, secondary and others outcomes.

5.2.1 Primary analysis.

Effects on primary outcomes

To investigate the overall effect of the resistance exercise intervention on the change from baseline to post-intervention in the executive function composite score, cLMM approach with repeated measures over time (baseline, 12-week and 24-week outcomes) will be used using an intention-to-treat principle. This statistical linear mixed model will include both a random intercept for each participant and a group-by- time interaction as a fixed effect.

5.2.2. Secondary analyses

Effects on secondary outcomes

To investigate the effect of the resistance exercise intervention on the secondary outcomes (other cognitive outcomes, brain structure and function and AB), the same analyses as the primary analysis using an intention-to-treat principle and cLMM analysis will be performed. The preprocessing analysis of brain structure and function measured with MRI will be carried out as secondary analyses using different programs (Freesurfer, FSL, CON, fMRIprep).

Effects on other outcomes

To investigate the effect of the resistance exercise intervention on other outcomes (i.e., biological biomarkers, physical function, muscular strength, physical activity, Anthropometric, body composition and blood pressure and Psychosocial, mental health and other questionnaires*)* the same analyses as the primary analysis using an intention-to-treat principle and cLMM analysis will be performed.

Furthermore, analyses per protocol will be followed of the executive function variables (ie., Executive function score, Trail making test, Digit symbol substitution score, Dimensional change card sort test and Spatial working memory).

Moderators and mediators

In addition, mediation and moderation analysis will be performed following AGReMA (A Guideline for Reporting Mediation Analyses) recommendations (46), using the group as independent variable (intervention group vs control group), executive function as outcome, and mediating and moderating variables.

Moderators include will be sex, age, education, apolipoprotein E (APOE) status, and Aβ status. Additionally, baseline data of various cognitive measures, including the composite executive function score, Trail Making Test, Digit Symbol Substitution Score, Dimensional Change Card Sort Test, and Spatial Working Memory, will be incorporated. The baseline values of strength measures and mental health variables, specifically depression and self-esteem, will be also analyzed. We will also assess the potential mediators, aiming to understand the mechanisms through which the intervention could exert its effects. No interim analysis will be performed.

5.2.3 Sensitivity analyses

A sensitivity analysis will be performed using an intention-to-treat principle with all randomly assigned participants, but excluding participants who encountered adverse events that may have impacted the primary outcomes. Adverse events considered for exclusion is conducted to identify and categorize adverse events based on predefined criteria using common terminology criteria for adverse events (i.e., mild, moderate, severe) (47).

5.3 Missing data

The number of missing data will be reported and patterns of missing data will be explored. Based on previous experience, we expect that missing data will be assumed as missing at random. Therefore, the linear mixed model analyses will handle our missing data. However,

once the data processing is finalized, we reconsider this expectation. In case we believe this assumption does not hold, we will explore appropriate measures for the data analyses.

5.4 Additional analyses/Compensatory analyses

A set of additional analyses will be performed to confirm the robustness of main the analyses. We will perform a 1-dimension curve analysis using an Statistical Parametric Mapping 1 package (SPM1D) available for MATLAB (http://www.spm1d.org) (33) to study whether acceleration values (i.e., expressed as ENMO [mg]) identify a significant increase in physical activity during the exercise program in comparison with the physical activity pattern at baseline for the control and exercise groups.

In-depth analysis of intervention compliance is going to be conducted to gauge the intensity as exploratory analysis. Changes in others outcomes will be also analyzed using a similar protocol as described by '5.2 Analysis methods' by following the intention to treat and per protocol principles. Furthermore, cross-sectional analyses will be performed using the baseline data of the RCT.

5.5 Harms

Any situation that may occur, such as injury, emergency or scheduled surgery was reported as adverse events categorize based on predefined criteria using common terminology criteria (i.e., mild, moderate, severe) (6). Sensitivity analyses will be done with all randomly assigned study participants who encountered adverse events that may have impacted the primary outcomes of the analyses.

5.6 Statistical software

The analyses of the primary outcomes will be performed using R. For the main analyses, we will use the 'LMMstar', lmer, nlme and lavaan packages. The use of specific packages will be reported in each manuscript. In addition, specific software's are used for secondary brain outcomes.

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